Fundamentals of Vector

Comparison of Vector and Scalar

Parameters	Scalar	Vector
Magnitude & Direction	❖ Scalar Quantities have only magnitude and It has no directions.	❖ Vector Quantities have magnitude as well as directions.
Dimensions	❖ Scalar Quantities have only One Dimension.	❖ Vector Quantities have more than one Dimension.
Division Operation	❖ Scalar Quantities can be divided by another Scalar Quantity.	❖ Vector Quantities can not be divided by another Vector Quantity.
Representation	❖ Number (Magnitude) and Unit are only required to represent scalar quantities. ❖ Example: T = 290 K	❖ Number (Magnitude), Direction (Unit Vector), and Unit are required to represent Vector quantities. ❖ Example: $\vec{E} = 290 \times 10^{-9}\ (\hat{i} + \hat{j})/\sqrt{2}\ V/m$
Example	❖ Mass, Temperature, Time etc.	❖ Velocity, Force, Electric Field, Magnetic Field etc.

Engineering Funda

Basics of Vector

- Representation of Vector

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$$\vec{A} = (A_x, A_y, A_z)$$

- Magnitude of Vector

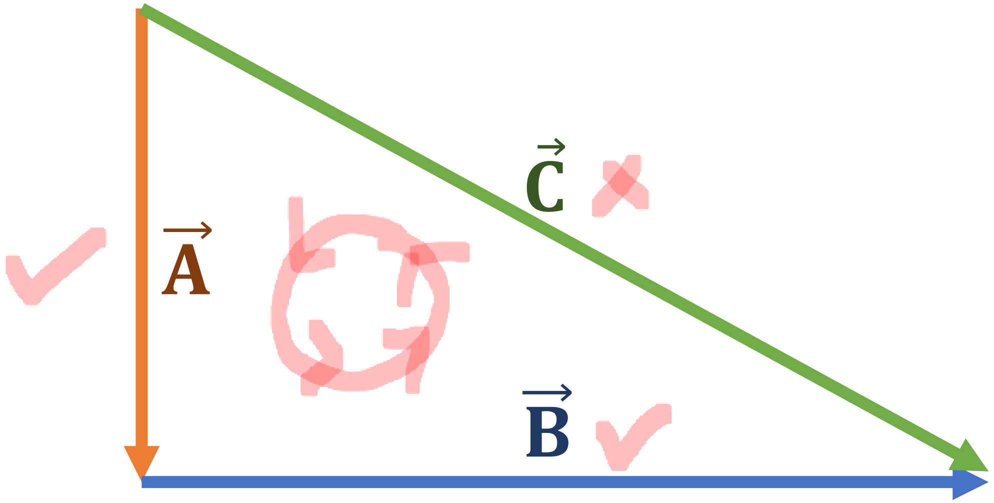
$$A = |\vec{A}| = \sqrt{{A_x}^2 + {A_y}^2 + {A_z}^2}$$

- Unit Vector

$$\widehat{i_A} = \frac{\vec{A}}{|\vec{A}|}$$

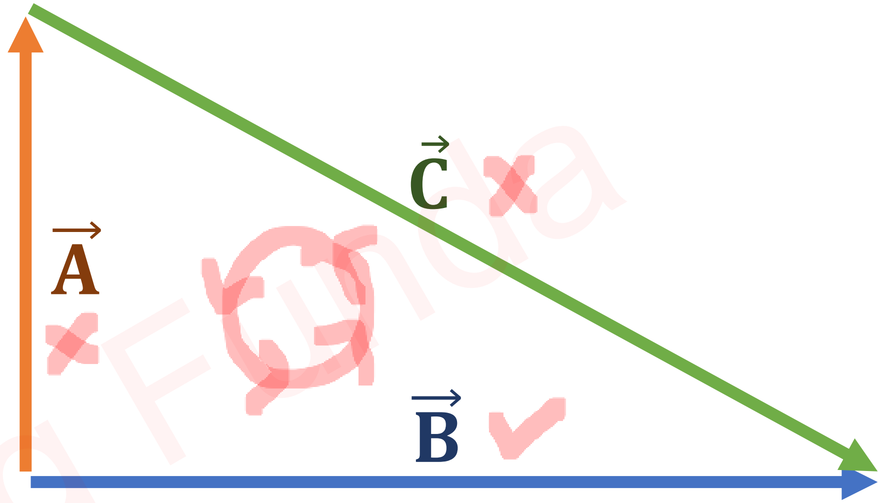
Engineering Funda

Law of Triangle



$\Rightarrow \vec{A} + \vec{B} - \vec{C} = 0$

$\Rightarrow \vec{C} = \vec{A} + \vec{B}$

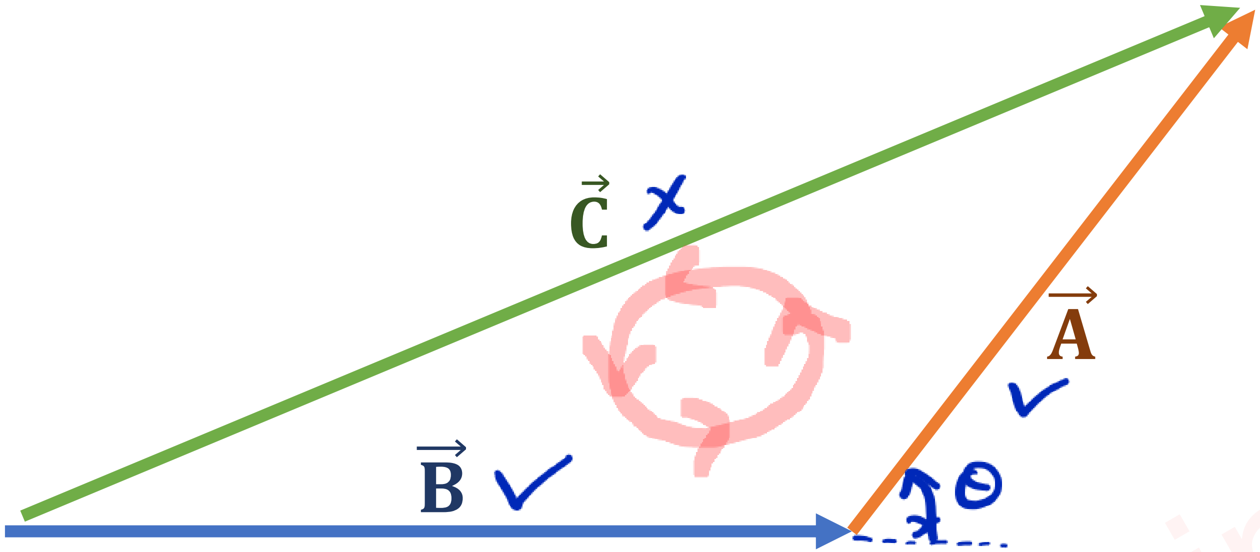


$\Rightarrow -\vec{A} - \vec{C} + \vec{B} = 0$

$\Rightarrow \vec{B} = \vec{A} + \vec{C}$

Engineering Funda

Resultant of Two Vectors



$$\rightarrow \quad \vec{A}+\vec{B}-\vec{C}=0$$

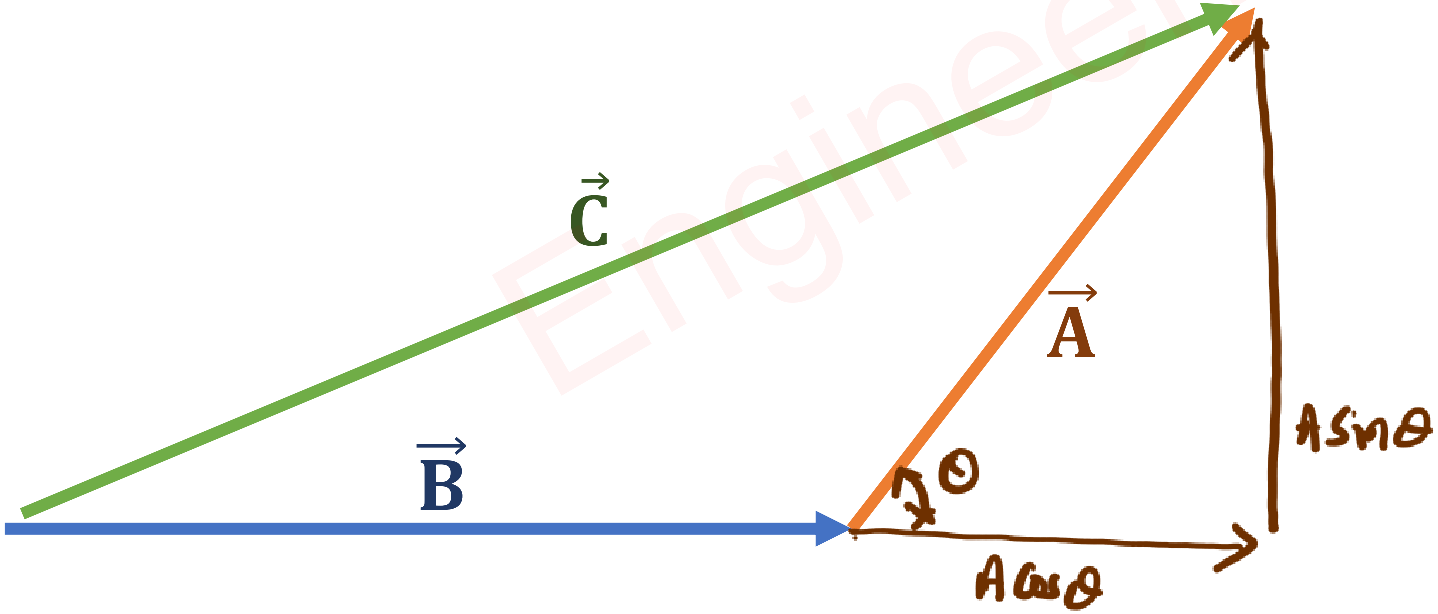
$$\Rightarrow \vec{C}=\vec{A}+\vec{B}$$

$$\Rightarrow \vec{C}\cdot\vec{C}=(\vec{A}+\vec{B})\cdot(\vec{A}+\vec{B})$$

$$\Rightarrow C^2=A^2+B^2+2\vec{A}\cdot\vec{B}$$

$$\Rightarrow C=\sqrt{A^2+B^2+2AB\cos\theta}$$

↝ From Triangle



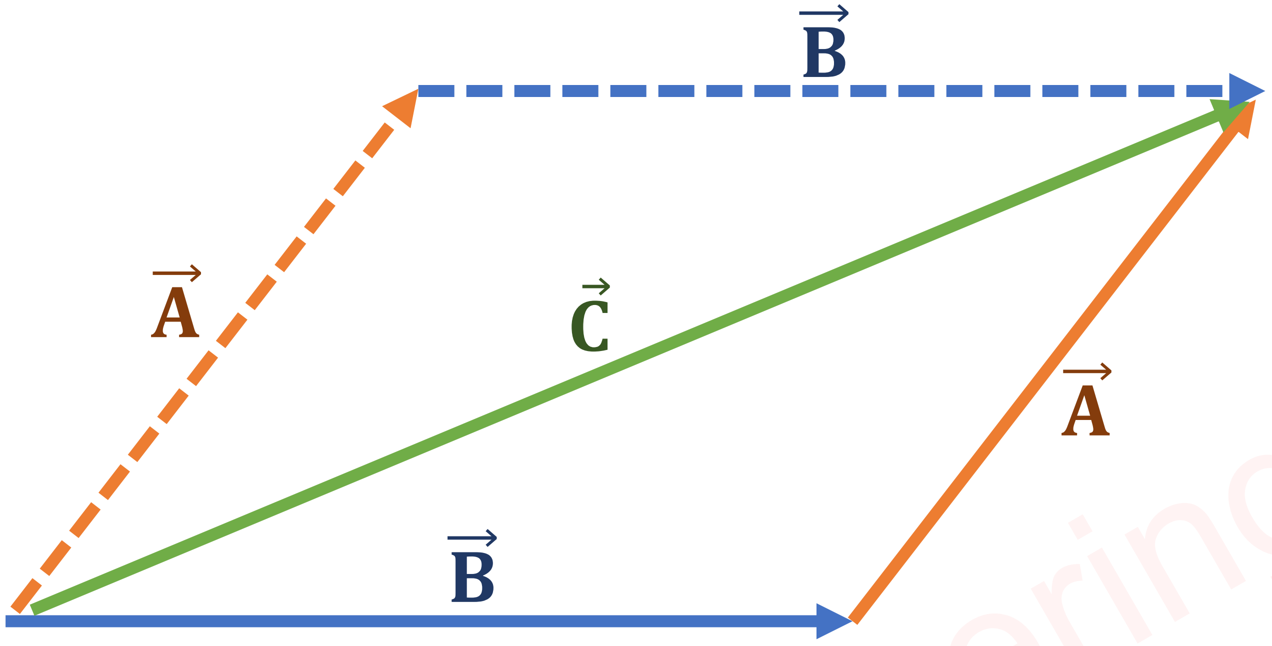
$$\Rightarrow C^2=A^2\sin^2\theta+(B+A\cos\theta)^2$$

$$\Rightarrow C=\sqrt{A^2(\sin^2\theta+\cos^2\theta)+B^2+2AB\cos\theta}$$

$$\Rightarrow C=\sqrt{A^2+B^2+2AB\cos\theta}$$

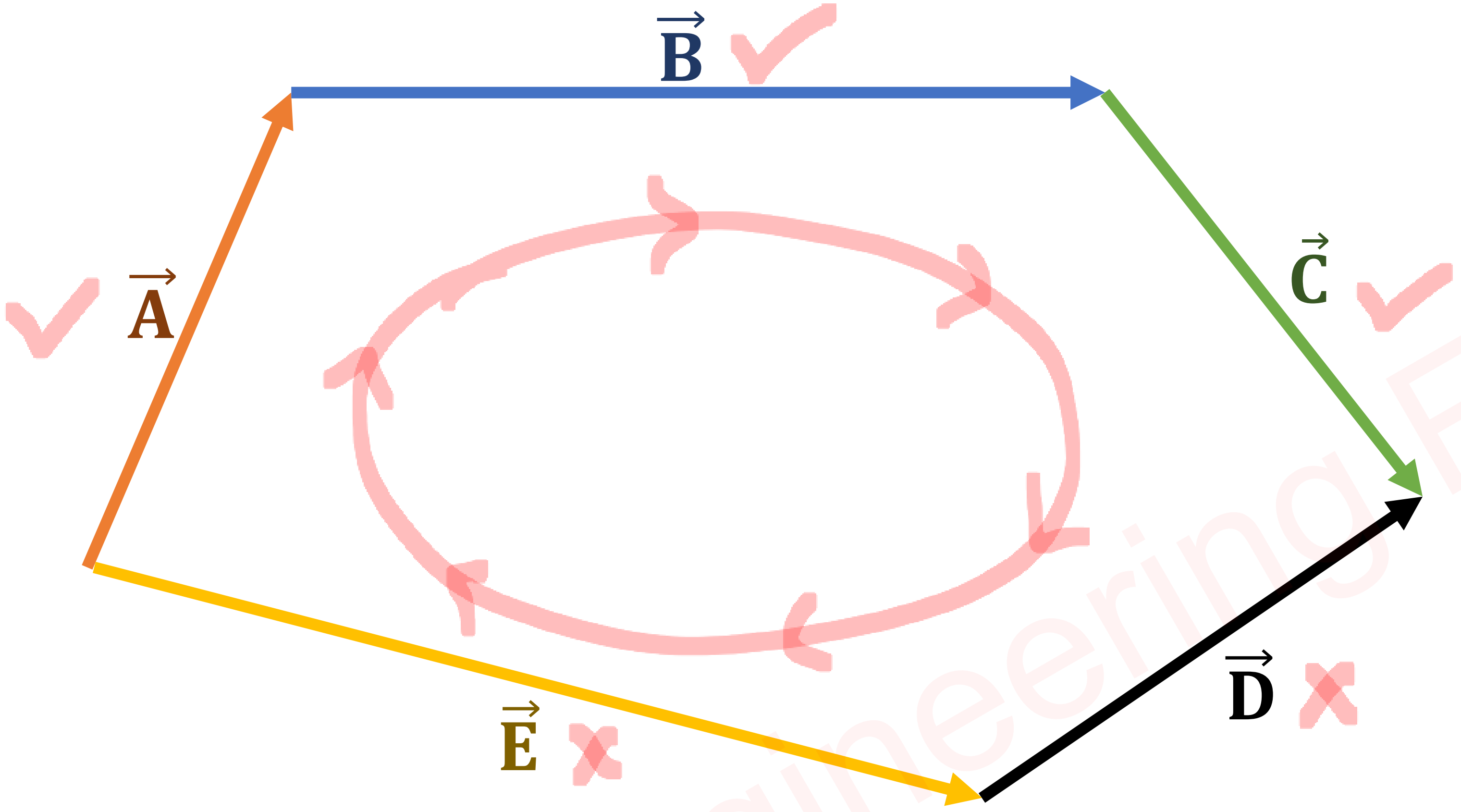
Engineering Funda

Law of Parallelogram



Engineering Funda

Law of Polygon



$$\vec{A} + \vec{B} + \vec{C} - \vec{D} - \vec{E} = 0$$

$$\Rightarrow \vec{A} + \vec{B} + \vec{C} = \vec{D} + \vec{E}$$

Engineering Funda

Vector Multiplication

Outlines

- Dot Product
- Cross Product
- Scaler Multiplication
- Properties of Dot Product
- Properties of Cross Product

Engineering Funda

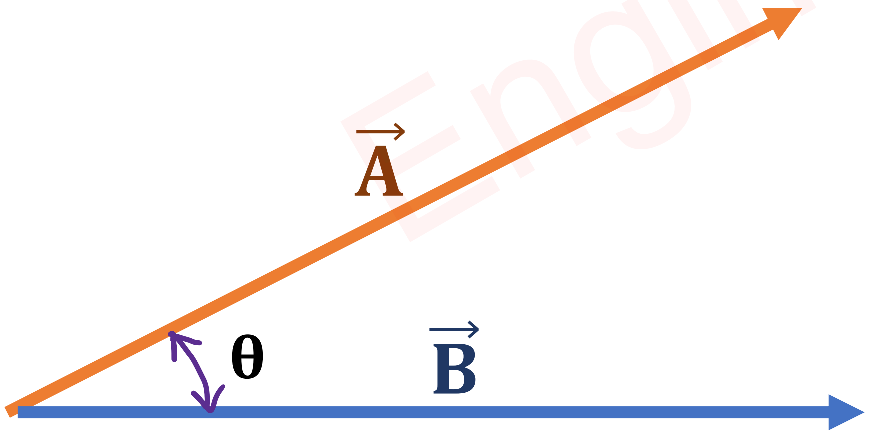
Dot Product

- Dot Product is a Scalar Product of Two vectors. So, the Resultant of the Dot Product is a Scalar Quantity.
- If two vectors are given by $\vec{A}$ and $\vec{B}$,

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$
$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

- Then Dot Product of $\vec{A}$ and $\vec{B}$ is given by:

$$\Rightarrow Y = \vec{A}.\vec{B} = |\vec{A}||\vec{B}| \cos\theta$$
$$\Rightarrow Y = \vec{A}.\vec{B} = A_xB_x + A_yB_y + A_zB_z$$



Engineering Funda

Cross Product

- Cross Product is a Vector Product of Two vectors. So, the result of the Cross-Product is a Vector Quantity.
- If two vectors are given by $\vec{A}$ and $\vec{B}$,

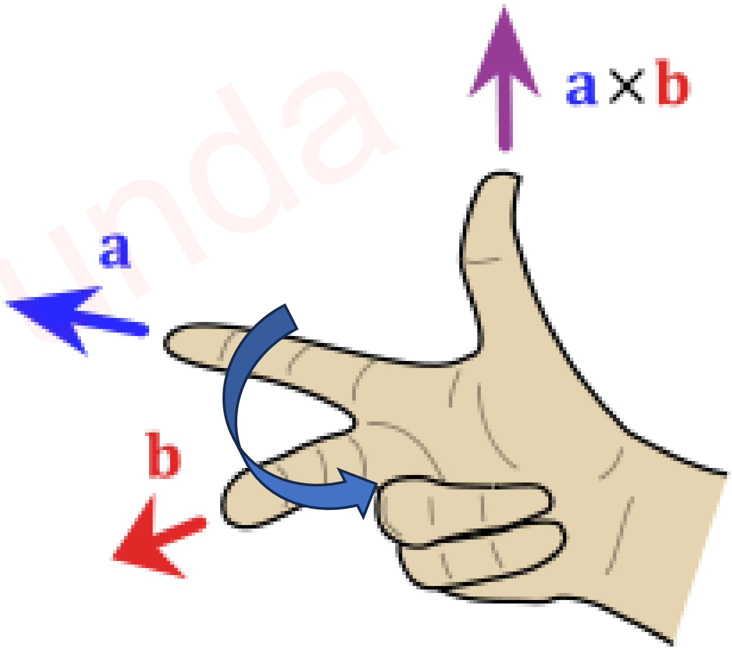
$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$
$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

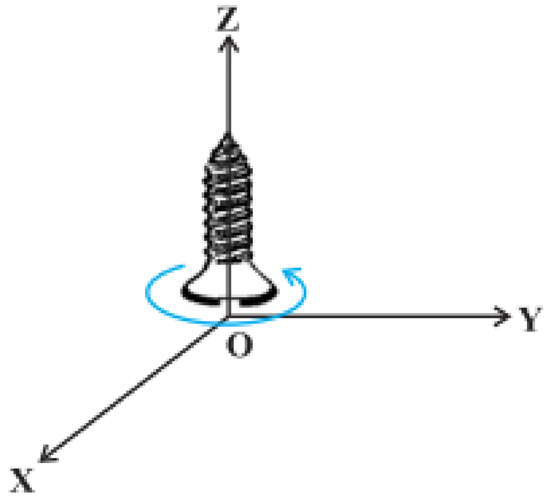
- Then Cross Product of $\vec{A}$ and $\vec{B}$ is given by:

$$\Rightarrow Y = \left|\vec{A} \times \vec{B}\right| = \left|\vec{A}\right|\left|\vec{B}\right| \sin\theta$$

$$\Rightarrow \vec{Y} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

- Direction of Cross Product is as per Right Hand Thumb Rule.
- Cross Product is always perpendicular to given two vectors.





Engineering Funda

Scaler Multiplication

- If The vector is given $\vec{A}$,

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

- Then scaler Multiplication is given by:

$$\Rightarrow \vec{Y} = k\vec{A} = kA_x\hat{i} + kA_y\hat{j} + kA_z\hat{k}$$

- After Scaler Multiplication, The Direction of the vector doesn’t change.
- After Scaler Multiplication, The Direction of $\vec{Y}$ remains in the direction of $\vec{A}$.

Engineering Funda

Properties of Dot Product and Cross Product

Dot Product	Cross Product
❖ $\vec{A}.\vec{B} = \vec{B}.\vec{A}$	❖ $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$
❖ $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$	❖ $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$
❖ $\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i} = 0$	❖ $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$

$\hat{i} \quad \hat{j} \quad \hat{k} \quad \hat{i} \quad \hat{j} \quad \hat{k}$

$\xrightarrow{\hspace{6em}}$ +ve

example : $\hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{k} = \hat{i}$

$\xleftarrow{\hspace{6em}}$ -ve

example : $\hat{j} \times \hat{i} = -\hat{k}$

$\hat{k} \times \hat{j} = -\hat{i}$

Engineering Funda

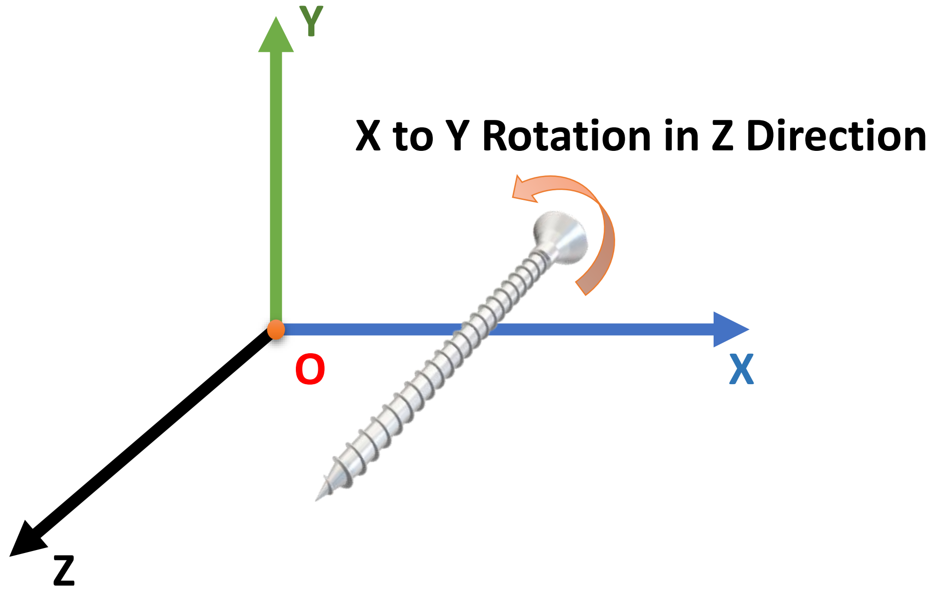
Cartesian Coordinate System

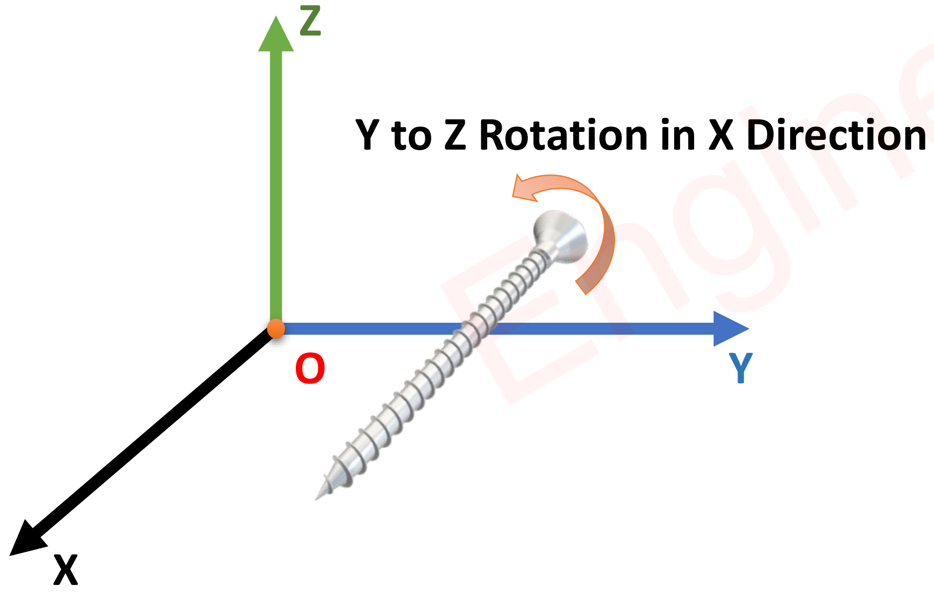
Outlines

- Direction Identification in Cartesian Coordinate System
- Representation of Cartesian Coordinate System
- Projection of Vector
- Solved Problem of Projection of Vector

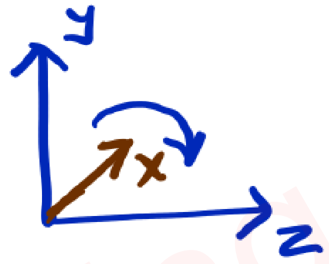
Engineering Funda

Direction Identification in Cartesian Coordinate System



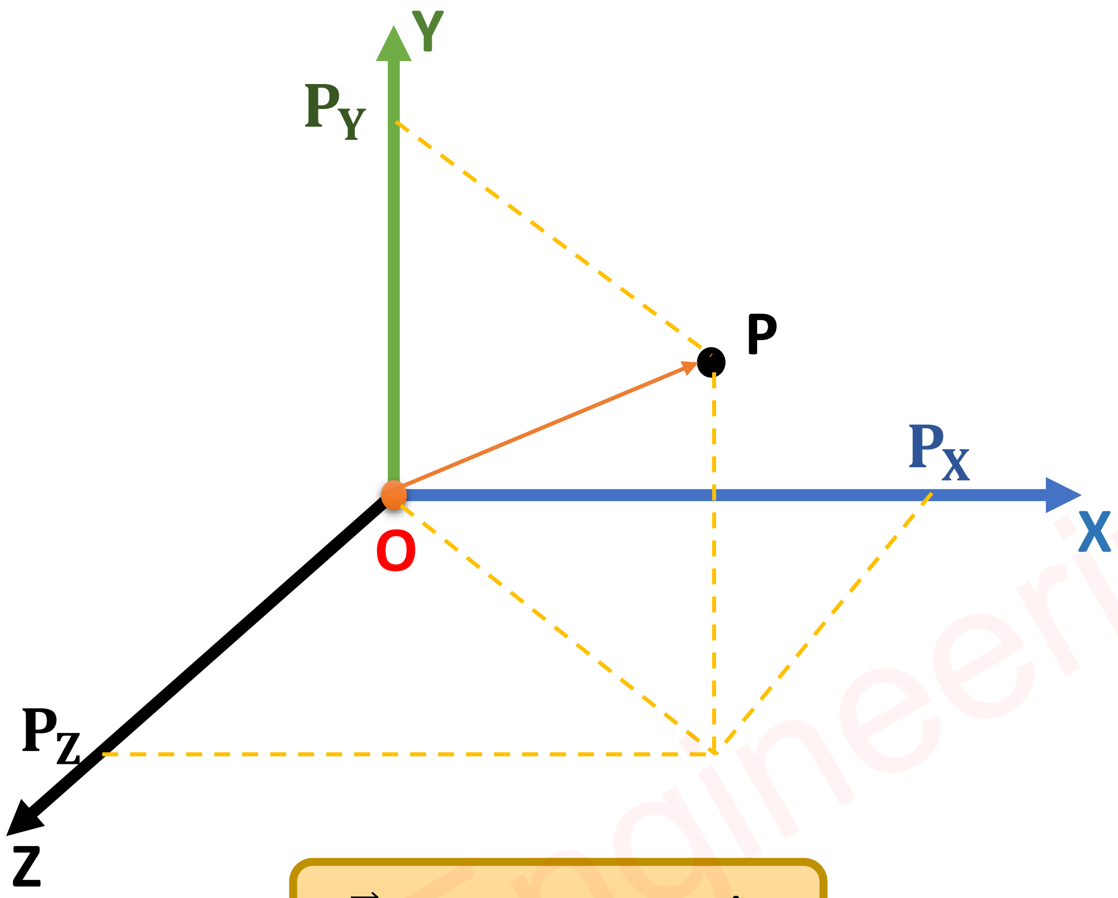


- ❑ Direction of Unknown Axis will be as per XYZXYZ sequence.
- ❑ Example: Z to X rotation is in the Y Direction.



Engineering Funda

Representation of Cartesian Coordinate System



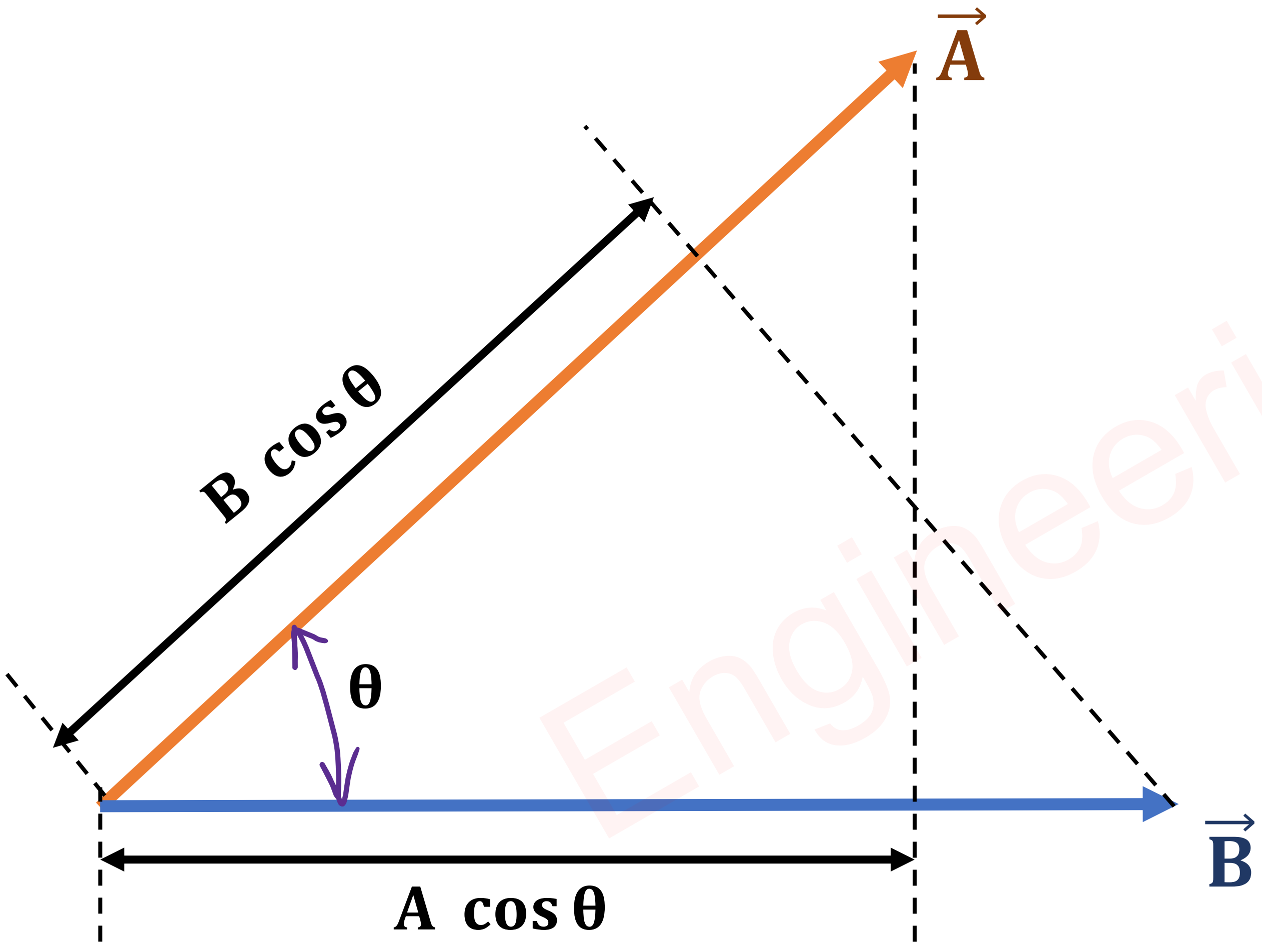
$$\vec{P} = P_x\hat{i} + P_y\hat{j} + P_z\hat{k}$$

$$P = |\vec{P}| = \sqrt{{P_x}^2 + {P_y}^2 + {P_z}^2}$$

Engineering Funda

Projection of Vector

- The Projection of Vector $\vec{A}$ on $\vec{B}$ is $A \cos\theta$, Where θ is Angle between $\vec{A}$ and $\vec{B}$.
- The Projection of Vector $\vec{B}$ on $\vec{A}$ is $B \cos\theta$, Where θ is Angle between $\vec{A}$ and $\vec{B}$.



$$\text{Cos}\,\theta = \frac{\vec{A}.\vec{B}}{AB}$$

Engineering Funda

Solved Problem of Projection of Vector

- Find The Projection of Vector $\vec{A}$ on $\vec{B}$ and Projection of $\vec{B}$ on $\vec{A}$.

$\underline{\vec{A} \text{ on } \vec{B}}$: $A\cos\theta$ $\quad$ $\underline{\vec{B} \text{ on } \vec{A}}$: $B\cos\theta$

$$\vec{A} = 3\hat{i} + 2\hat{j} - \hat{k}$$
$$\vec{B} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$\rightsquigarrow \vec{A}\cdot\vec{B} = AB\cos\theta$

$\Rightarrow A\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{B}$

$\Rightarrow B\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{A}$

$\rightsquigarrow A = |\vec{A}| = \sqrt{3^2 + 2^2 + (-1)^2} = \sqrt{14}$

$B = |\vec{B}| = \sqrt{2^2 + 3^2 + 6^2} = 7$

$\vec{A}\cdot\vec{B} = 3\times2 + 2\times3 - 6$

$= 6$

$\rightsquigarrow A\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{B} = \dfrac{6}{7}$

$\rightsquigarrow B\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{A} = \dfrac{6}{\sqrt{14}}$

Engineering Funda

Gradient, Divergence & Curl

Outlines

- Basics of Gradient
- Solved Example of Gradient
- Basics of Divergence
- Solved Example of Divergence
- Basics of Curl
- Example of Curl

Engineering Funda

Basics of Gradient

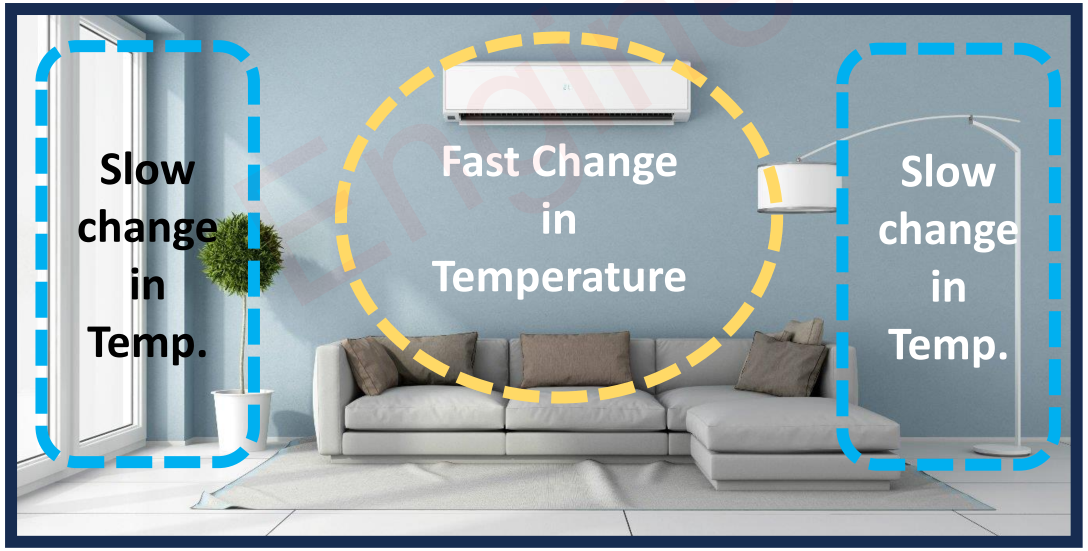
- Del Operator $(\vec{\nabla})$ is a vector operator, that we use in Gradient, Divergence and Curl.

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

- Gradient is applied on scaler quantity. A Gradient of function (F) can be calculated by:

$$\vec{\nabla}F = \frac{\partial F}{\partial x}\hat{i} + \frac{\partial F}{\partial y}\hat{j} + \frac{\partial F}{\partial z}\hat{k}$$

- So, After the Gradient of the function, we get vector quantity.
- The Gradient explains the function variation in x, y, and z directions.
- Example: Room Temperature {Function} change due to AC.



Engineering Funda

Solved Example of Gradient

- Find the Gradient of function F at point (1, 2, 3).

$$F = xy^2 + z^3y$$

$$\text{Grad (F)} = \vec{\nabla}F = \frac{\partial F}{\partial x}\hat{\imath} + \frac{\partial F}{\partial y}\hat{\jmath} + \frac{\partial F}{\partial z}\hat{k}$$

$$\text{Grad (F)} = \vec{\nabla}F = \frac{\partial(xy^2 + z^3y)}{\partial x}\hat{\imath} + \frac{\partial(xy^2 + z^3y)}{\partial y}\hat{\jmath} + \frac{\partial(xy^2 + z^3y)}{\partial z}\hat{k}$$

$$\text{Grad (F)} = \vec{\nabla}F = y^2\hat{\imath} + (2yx + z^3)\hat{\jmath} + 3z^2y\hat{k}$$

- At point (1, 2, 3).

$$\text{Grad (F)} = \vec{\nabla}F = 4\hat{\imath} + 31\hat{\jmath} + 54\hat{k}$$

Engineering Funda

Basics of Divergence

- Divergence is applied to Vector quantity. Divergence of Function is scaler Quantity.
- Divergence of function ($\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$) can be calculated by:

$$\vec{\nabla}.\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

- Mathematical Definition of Divergence is given by:

$$\Longrightarrow \vec{\nabla}.\vec{F} = \lim_{\Delta V \to 0} \frac{\oint \vec{F}.\overrightarrow{ds}}{\Delta V}$$

- By taking volume Integration:

$$\Longrightarrow \int \vec{\nabla}.\vec{F}\,dV = \oint \vec{F}.\overrightarrow{ds}$$

- Divergence Theorem explains the relation between Volume integration and Surface Integration.
- Divergence explains the overall function variation in x, y, and z direction.

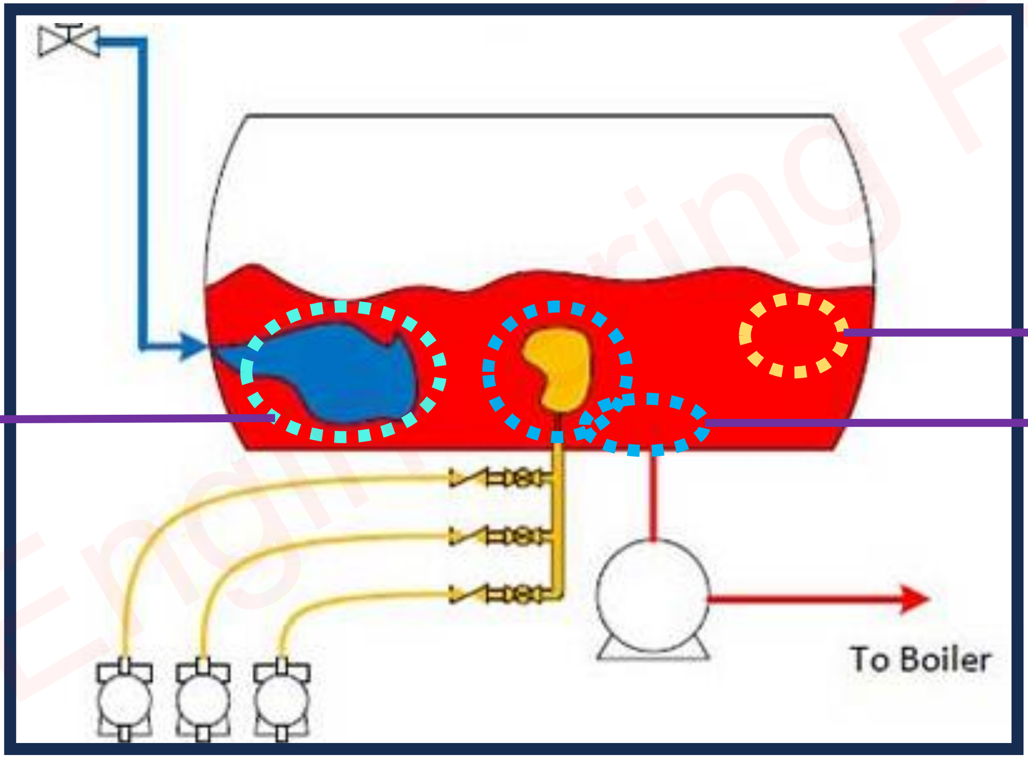
Engineering Funda

Example of Divergence

- ❑ Divergence explains the overall change of a function with respect to x, y, and Z coordinates.

$$\vec{\nabla}.\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

- ❑ Water is a function for given Tank.



Engineering Funda

Solved Example of Divergence

- Find the Divergence of function $\vec{F}$ at point (1, 2, 1).

$$\vec{F} = xy^2\hat{i} + y\hat{j} + xz\hat{k}$$

$$Div\left(\vec{F}\right) = \vec{\nabla}.\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$Div\left(\vec{F}\right) = \vec{\nabla}.\vec{F} = \frac{\partial xy^2}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial xz}{\partial z}$$

$$Div\left(\vec{F}\right) = \vec{\nabla}.\vec{F} = y^2 + 1 + x$$

- At point (1, 2, 1).

$$Div\left(\vec{F}\right) = \vec{\nabla}.\vec{F} = 2^2 + 1 + 1 = 6$$

Engineering Funda

Basics of Curl

- Curl is applied to Vector quantity. Curl of Function is Vector Quantity.
- Curl of function ($\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$) can be calculated by:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

- Curl of vector explains Stokes Theorem

$$\Rightarrow \vec{\nabla} \times \vec{F} = \lim_{\Delta S \to 0} \frac{\oint \vec{F}.\overrightarrow{dl}}{\Delta S}$$

- By taking Surface Integration:

$$\Rightarrow \int \vec{\nabla} \times \vec{F}\, dS = \oint \vec{F}.\overrightarrow{dl}$$

- Stokes Theorem explains the relation between Surface integration and Line Integration.
- Curl explains variation with position and direction.

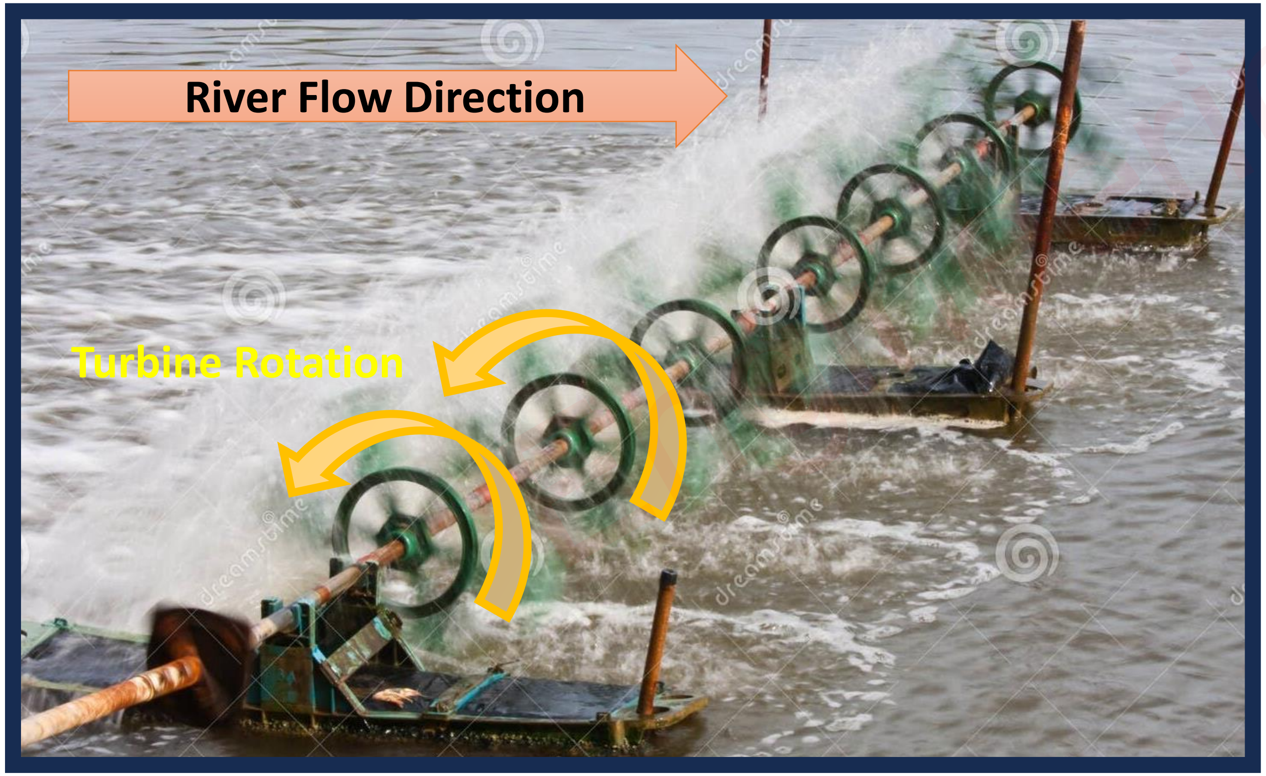
Engineering Funda

Example of Curl

- Curl explains variation with position and direction.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

- Rotation of Turbine is a function for given River Flow.



Gauss Divergence Theorem

Outlines

- Basics of Gauss Divergence Theorem
- Statement of Gauss Divergence Theorem
- Proof of Gauss Divergence Theorem
- Physical Significance of Gauss Divergence Theorem
- Applications of Gauss Divergence Theorem

Engineering Funda

Basics of Gauss Divergence Theorem

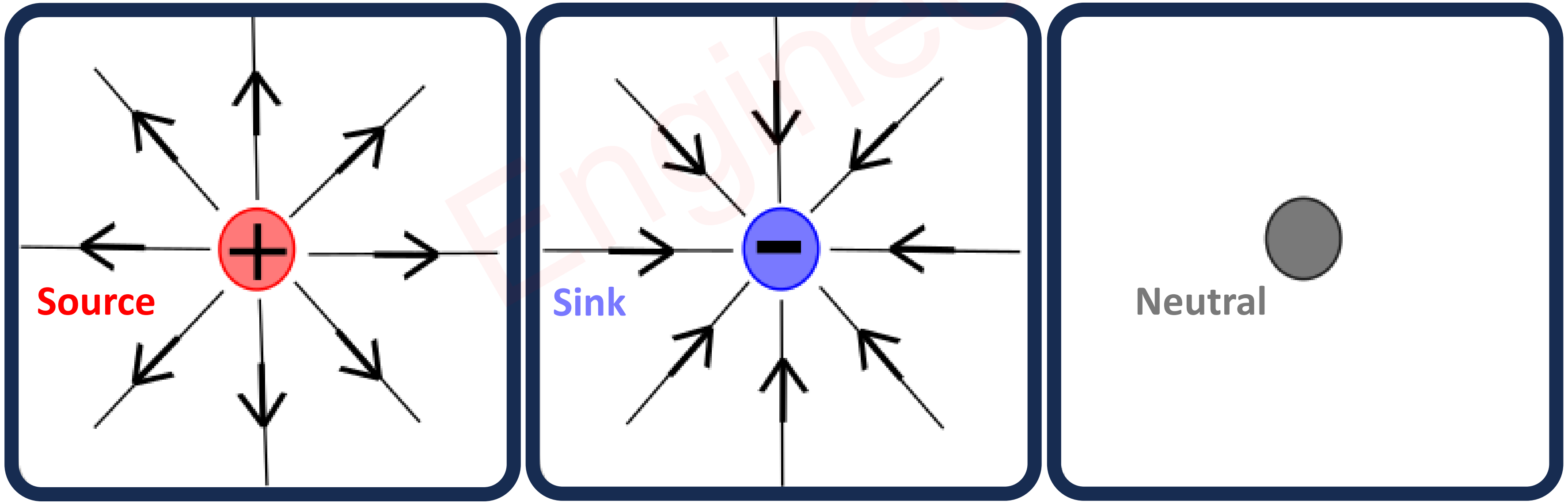
- Gauss Divergence Theorem explains the relationship between Volume Integration & Surface Integration.

$$\Rightarrow \int \vec{\nabla}.\vec{F}\, dV = \oint \vec{F}.\overrightarrow{ds}$$

- Gauss Divergence Theorem explains the rate of change of a function with respect to Coordinates x, y and z.

$$\Rightarrow \vec{\nabla}.\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

- Gauss Divergence Theorem is also used to see the location of Source and Sink.
- Divergence is Flux Density, it explains how much flux is entering or leaving from the source or Sink.



Engineering Funda

Proof of Gauss Divergence Theorem

- Divergence of function $\left(\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}\right)$ is given by:

$$\Rightarrow \vec{\nabla}.\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

- Divergence of function $\left(\vec{F}\right)$ can also be calculated by:

$$\Rightarrow \vec{\nabla}.\vec{F} = \lim_{\Delta V \to 0} \frac{\oint \vec{F}.\overrightarrow{ds}}{\Delta V}$$

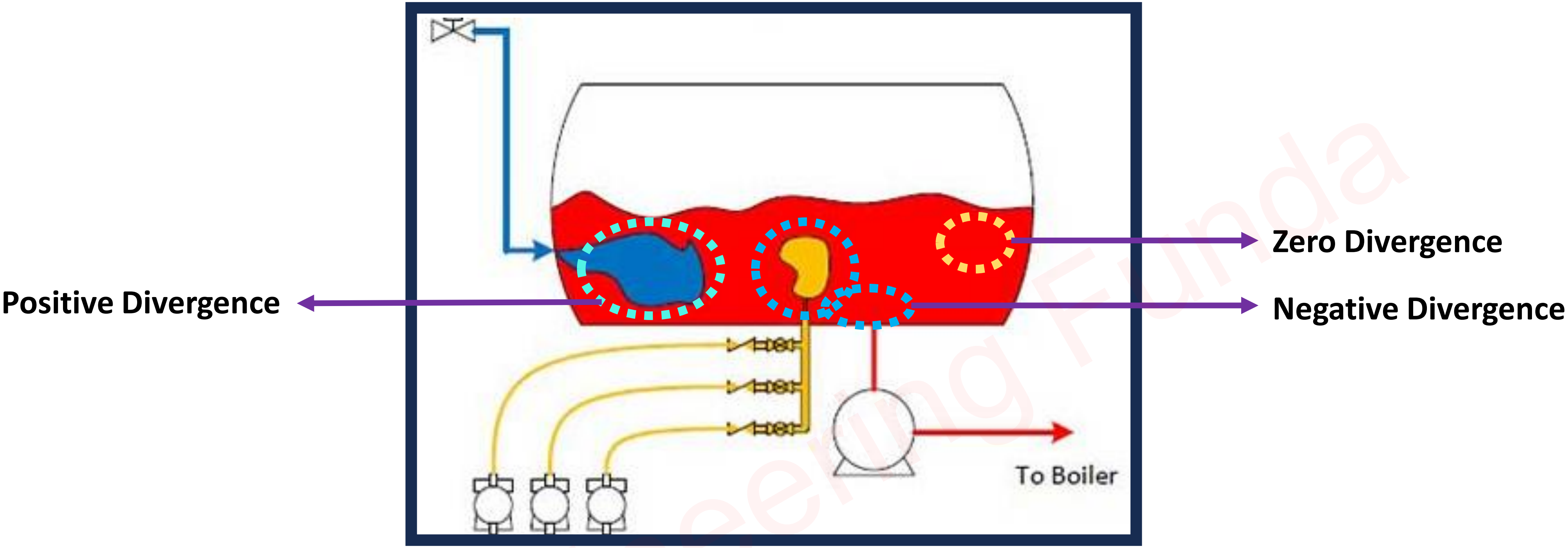
$$\Rightarrow \lim_{\Delta V \to 0} \vec{\nabla}.\vec{F}\,\Delta V = \lim_{\Delta V \to 0} \oint \vec{F}.\overrightarrow{ds}$$

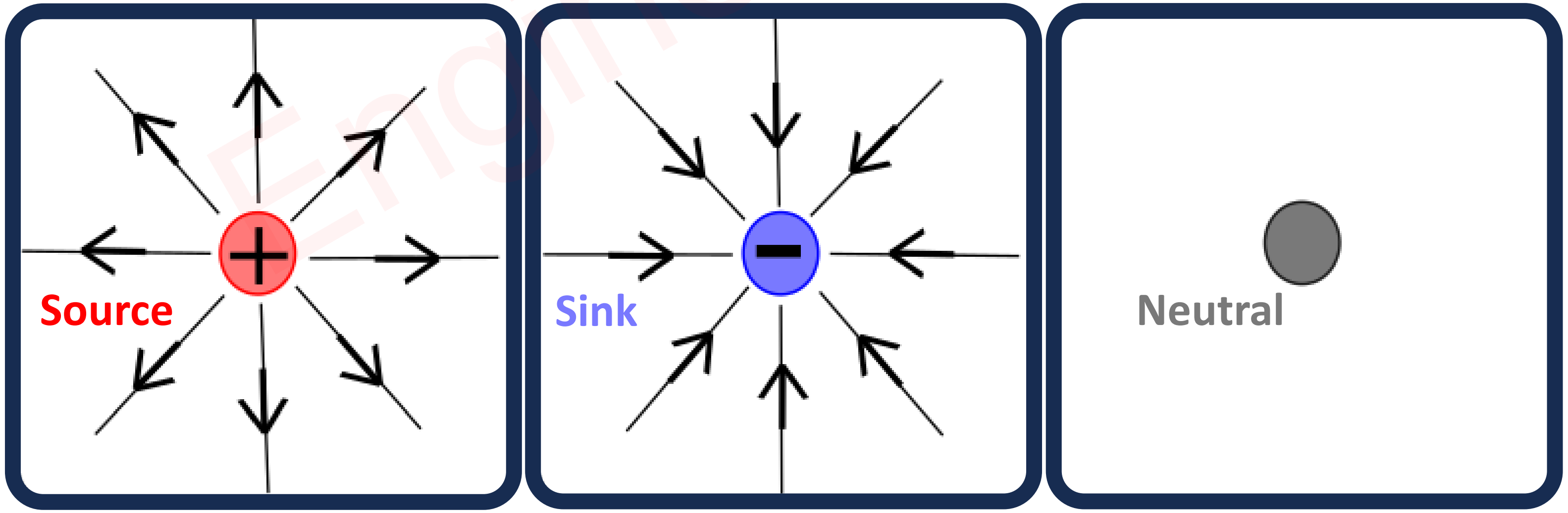
- So for total volume:

$$\Rightarrow \int \vec{\nabla}.\vec{F}\,dV = \oint \vec{F}.\overrightarrow{ds}$$

Engineering Funda

Physical Significance of Gauss Divergence Theorem





Engineering Funda

Applications of Gauss Divergence Theorem

- It is used in Applications of Fluid Mechanics.
- It is used in a flaw of fields. {Gravitational Fields, Electric Fields, Magnetic Fields etc.}
- It is used in Aerodynamics.
- It is used in Electromagnetics.

Engineering Funda

Stokes Theorem

Outlines

- Basics of Stokes Theorem
- Statement of Stokes Theorem
- Proof of Stokes Theorem
- Physical Significance of Stokes Theorem
- Applications of Stokes Theorem

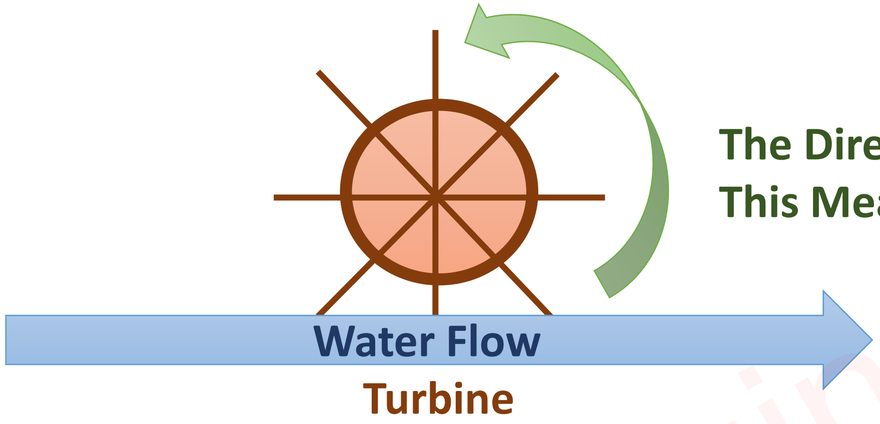
Engineering Funda

Basics of Stokes Theorem

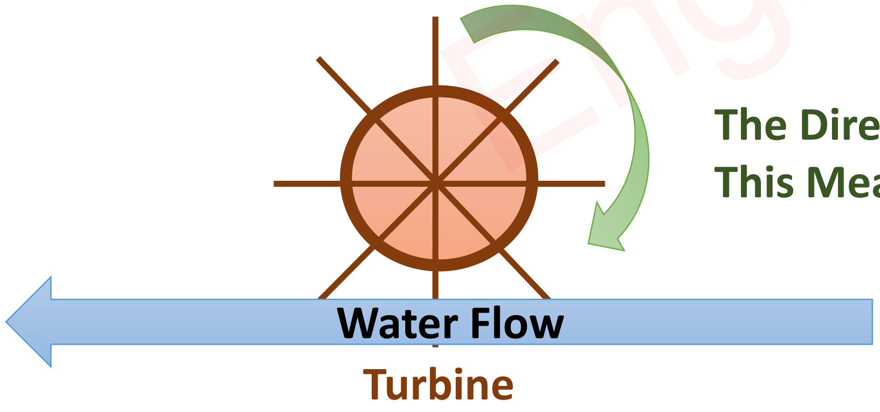
- Stokes Theorem explains the relationship between Line Integration & Surface Integration.

$$\Rightarrow \int \vec{\nabla} \times \vec{F}\, dS = \oint \vec{F}.\overrightarrow{dl}$$

- Stokes's Theorem is based on the Curl of Function.
- The Curl of Function explains the rotation of the body at different positions.



The Direction of Rotation in Anti Clockwise direction,
This Means the Curl of function is Positive.



The Direction of Rotation in a Clockwise direction,
This Means the Curl of function is Negative.

Engineering Funda

Proof of Stokes Theorem

- Curl of function $\left(\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}\right)$ is given by:

$$\Rightarrow \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

- Curl of function $\left(\vec{F}\right)$ can also be calculated by:

$$\Rightarrow \vec{\nabla} \times \vec{F} = \lim_{\Delta S \to 0} \frac{\oint \vec{F}.\vec{dl}}{\Delta S}$$

$$\Rightarrow \lim_{\Delta S \to 0} \vec{\nabla} \times \vec{F}\,\Delta S = \lim_{\Delta S \to 0} \oint \vec{F}.\vec{dl}$$

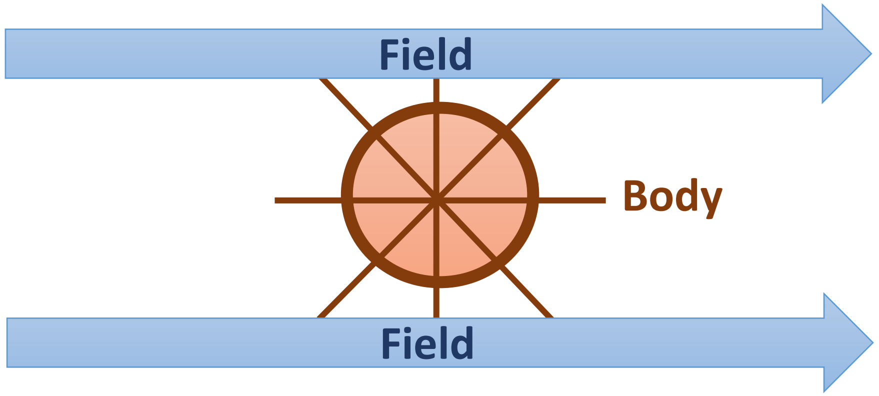
- For Total Surface:

$$\Rightarrow \int \vec{\nabla} \times \vec{F}\,dS = \oint \vec{F}.\vec{dl}$$

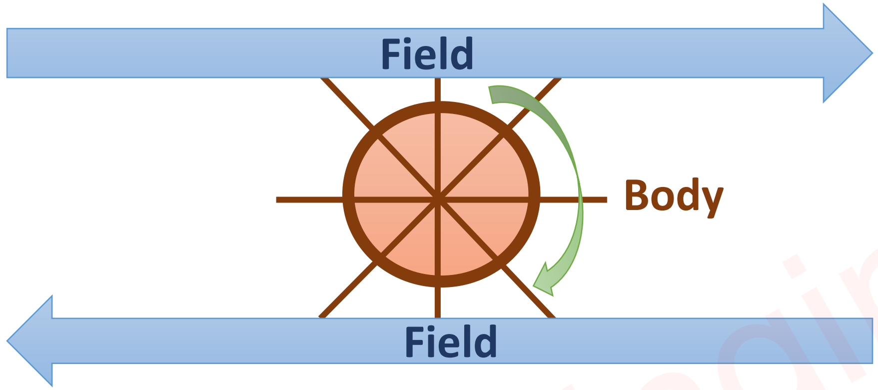
Engineering Funda

Physical Significance of Stokes Theorem

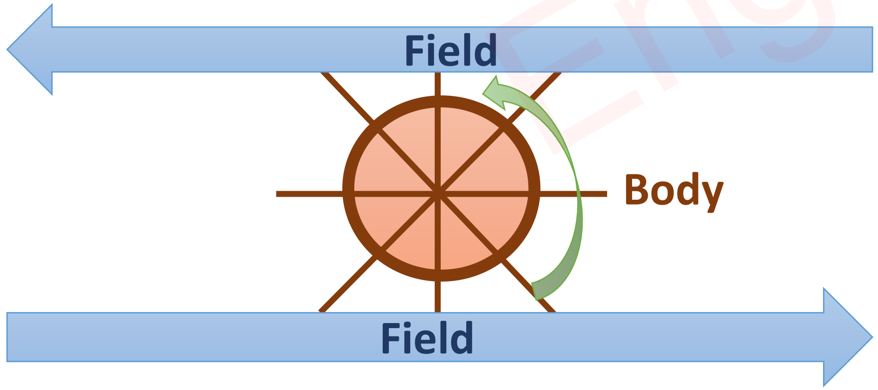
- Stokes Theorem explains whether given fields are Rotational or Irrotational.



Irrotational Field, means Curl of Function (Field) is Zero.



Rotational Field {Clockwise}, means Curl of Function (Field) is Negative.



Rotational Field {Anti Clockwise}, means Curl of Function (Field) is Positive.

Engineering Funda

Applications of Stocks Theorem

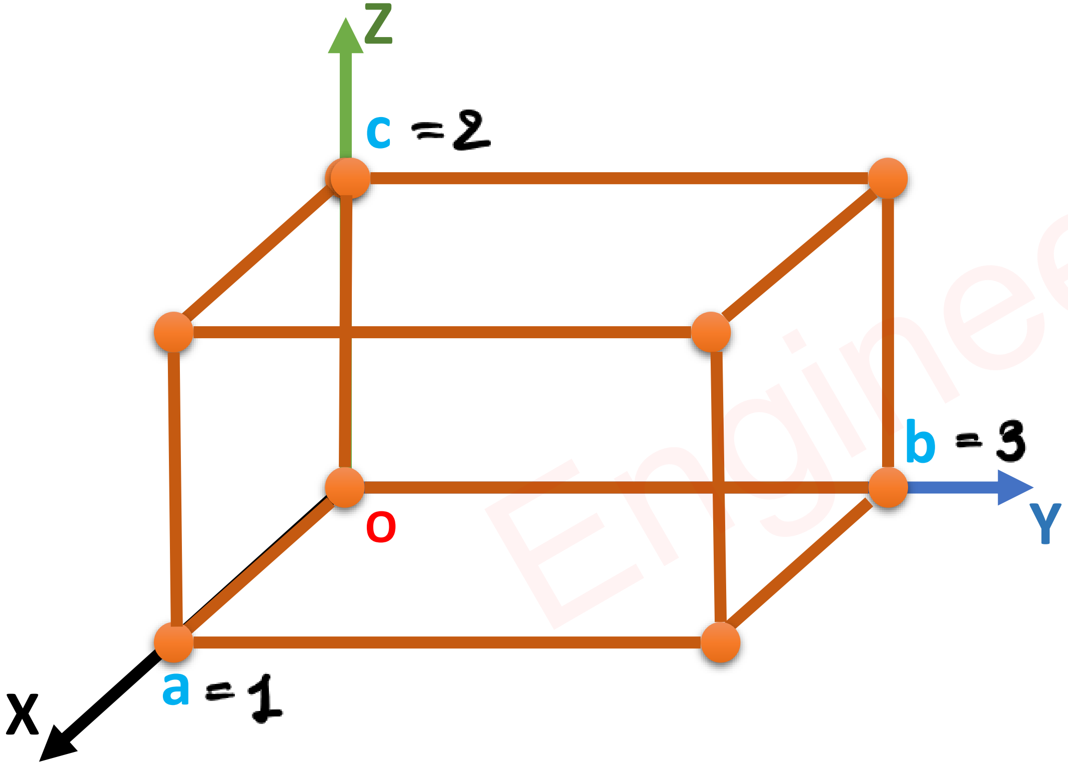
- ❑ It is used in Applications of Fluid Mechanics.
- ❑ It is used in a flaw of fields. {Gravitational Fields, Electric Fields, Magnetic Fields etc.}
- ❑ It is used in Aerodynamics.
- ❑ It is used in Electromagnetics.

Engineering Funda

Example of Divergence Theorem

❑ If $\vec{F} = xy\,\hat{i} + yz\,\hat{j} + zx\,\hat{k}$, a = 1, b = 3, c = 2, Then Find,

- $\int \vec{\nabla}.\vec{F}\ dV = \underline{\underline{18}}$ ✓
- $\int \vec{F}.\overrightarrow{ds} = ?\ \underline{\underline{18}}$



$$\vec{\nabla}.\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$= y + z + x$$

$x \in (0,1)$

$y \in (0,3)$

$z \in (0,2)$

$$\int \vec{\nabla}.\vec{F}\,dV = \int_0^2\int_0^3\int_0^1 (x+y+z)\,dx\,dy\,dz$$

$$= \int_0^2\int_0^3 \left(\frac{x^2}{2} + yx + zx\right)_0^1 dy\,dz$$

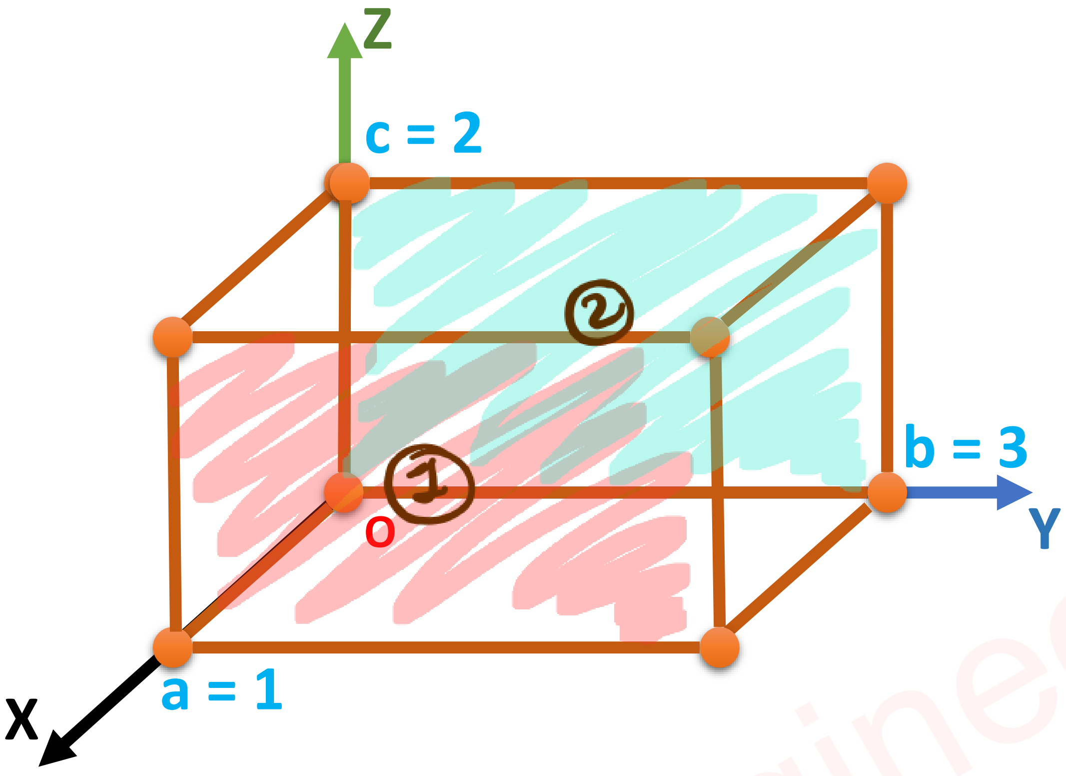
$$= \int_0^2\int_0^3 \left(\frac{1}{2} + y + z\right) dy\,dz$$

$$= \int_0^2 \left(\frac{1}{2}y + \frac{y^2}{2} + zy\right)_0^3 dz$$

$$= \int_0^2 (6 + 3z)\,dz$$

$$= \left(6z + 3\frac{z^2}{2}\right)_0^2 = 18$$

$$\vec{F} = xy\,\hat{i} + yz\,\hat{j} + zx\,\hat{k}$$



- $x = 1$

  $y \in (0,3)$

  $z \in (0,2)$

- $d\vec{S} = dy\,dz\,\hat{i}$

- $\displaystyle\int_{1} \vec{F}\cdot d\vec{S} = \int_0^2\int_0^3 xy\,dy\,dz$

  $\displaystyle = [x]\left[\frac{y^2}{2}\right]_0^3 [z]_0^2 = 9$

- $x = 0$

  $y \in (0,3)$

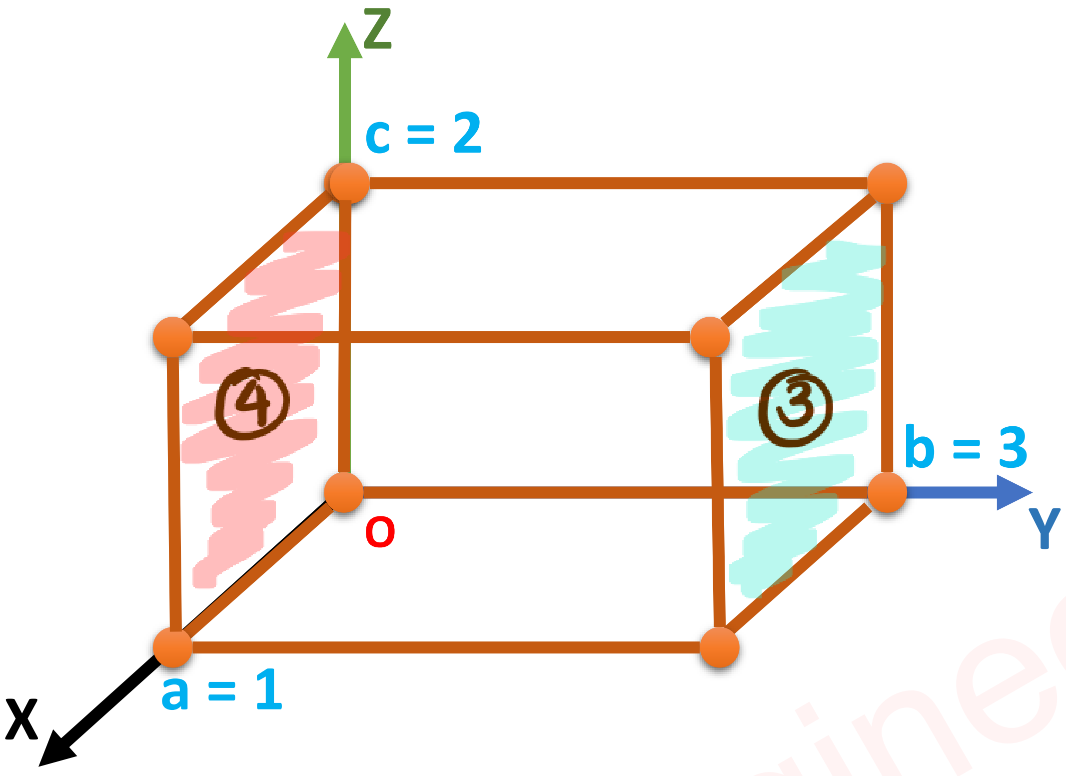
  $z \in (0,2)$

- $d\vec{S} = dy\,dz\,\hat{i}$

- $\displaystyle\int_{2} \vec{F}\cdot d\vec{S} = \int\int xy\,dy\,dz = 0$

Engineering Funda

$$\vec{F} = xy\,\hat{i} + yz\,\hat{j} + zx\,\hat{k}$$



$\rightsquigarrow$ $y = 3$

$x \in (0, 1)$

$z \in (0, 2)$

$\rightsquigarrow$ $d\vec{s} = dx\,dz\,\hat{j}$

$\rightsquigarrow$ $\displaystyle\int_{3} \vec{F}\cdot d\vec{s} = \int_0^2\int_0^1 yz\,dx\,dz$

$\displaystyle = (3)\left(\frac{z^2}{2}\right)_0^2 (x)_0^1 = 6$

$\rightsquigarrow$ $y = 0$

$x \in (0, 1)$

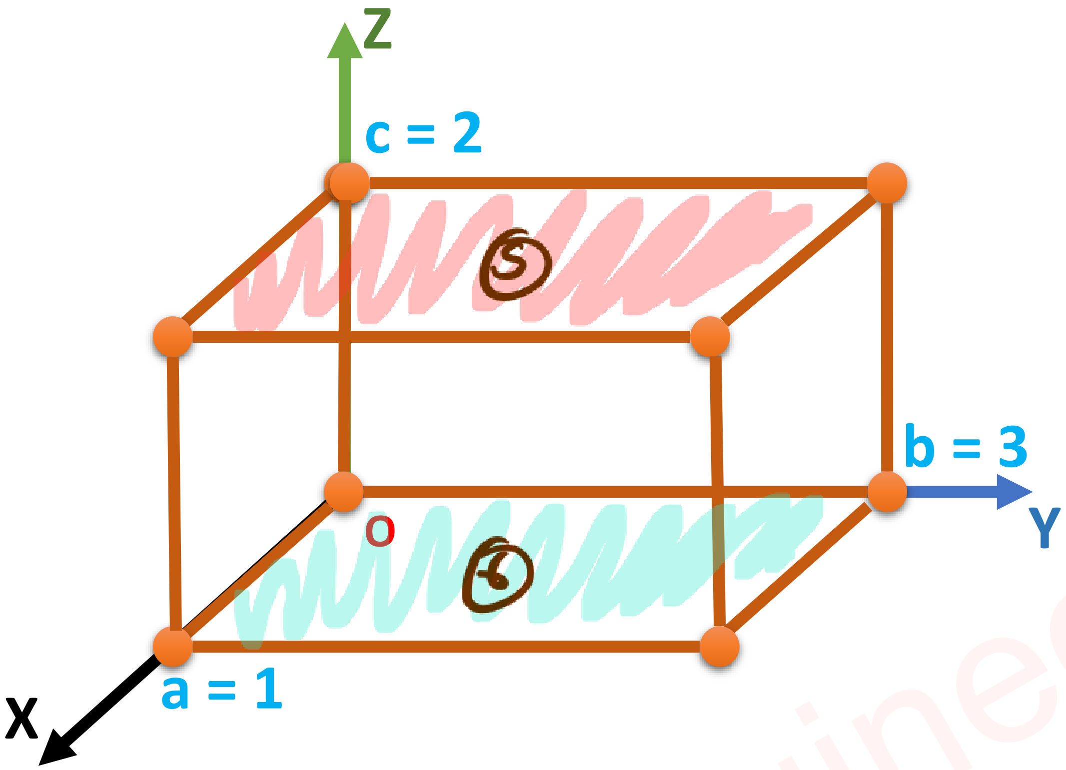
$z \in (0, 2)$

$\rightsquigarrow$ $d\vec{s} = dx\,dz\,\hat{j}$

$\rightsquigarrow$ $\displaystyle\int_{4} \vec{F}\cdot d\vec{s} = \int\int yz\,dx\,dz = 0$

Engineering Funda

$$\vec{F} = xy\,\hat{i} + yz\,\hat{j} + zx\,\hat{k}$$



$\sim z = 2$

$x \in (0,1)$

$y \in (0,3)$

$\sim d\vec{s} = dx\,dy\,\hat{k}$

$$\sim \int_{5} \vec{F}\cdot d\vec{s} = \int_0^3\int_0^1 2x\,dx\,dy$$

$$= 2\times\left(\frac{x^2}{2}\right)_0^1 (y)_0^3 = 3$$

$\sim z = 0$

$x \in (0,1)$

$y \in (0,3)$

$\sim d\vec{s} = dx\,dy\,\hat{k}$

$$\sim \int_{6} \vec{F}\cdot d\vec{s} = \int\int 2x\,dx\,dy = 0$$

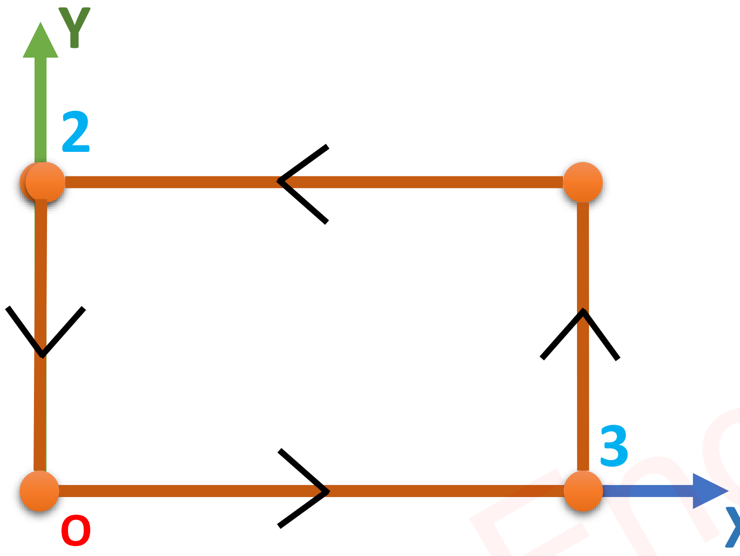
$$\sim \int \vec{F}\cdot d\vec{s} = 9+6+3$$

$$= \underline{\underline{18}}$$

Engineering Funda

Example of Stokes Theorem

- If $\vec{F} = x^2y\,\hat{i} + (x + y^2)\,\hat{j} + xy\,\hat{k}$, Then Find,
  - $\int \vec{\nabla} \times \vec{F}\ \overrightarrow{dS} = \cancel{?}\ \underline{-12}$
  - $\int \vec{F} . \overrightarrow{dl} = \cancel{?}\ \underline{-12}$



$$\sim \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y & x+y^2 & xy \end{vmatrix} = \hat{i}\left(\frac{\partial(xy)}{\partial y} - \frac{\partial(x+y^2)}{\partial z}\right) - \hat{j}\left(\frac{\partial(xy)}{\partial x} - \frac{\partial(x^2y)}{\partial z}\right) + \hat{k}\left(\frac{\partial(x+y^2)}{\partial x} - \frac{\partial(x^2y)}{\partial y}\right)$$

$$= \hat{i}(x) - \hat{j}(y) + \hat{k}(1-x^2)$$

$$\sim \vec{dS} = dx\,dy\,\hat{k}, \quad x \in (0,3), \quad y \in (0,2)$$

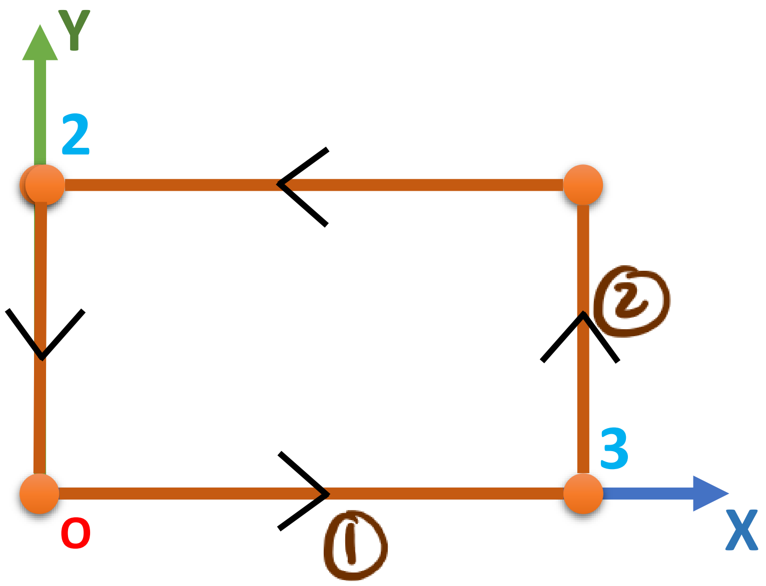
$$\sim \int \vec{\nabla} \times \vec{F}\ \vec{dS} = \int_0^2 \int_0^3 (1-x^2)\,dx\,dy$$

$$= \left(x - x^3/3\right)_0^3 \left(y\right)_0^2$$

$$= (3-9)(2)$$

$$= -12$$

$\vec{F} = x^2y\,\hat{i} + (x + y^2)\,\hat{j} + xy\,\hat{k}$



$\rightsquigarrow\ d\vec{l} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$

$\rightsquigarrow\ d\vec{l} = dx\,\hat{i}\ ,\ x \in (0,3)\ ,\ y = 0$

$\rightsquigarrow\ \int_{①} \vec{F}\,d\vec{l} = \int_0^3 x^2y\,dx = 0$

$\rightsquigarrow\ d\vec{l} = dy\,\hat{j}\ ,\ y \in (0,2)\ ,\ x = 3$

$\rightsquigarrow\ \int_{②} \vec{F}\cdot d\vec{l} = \int_0^2 (x+y^2)\,dy$

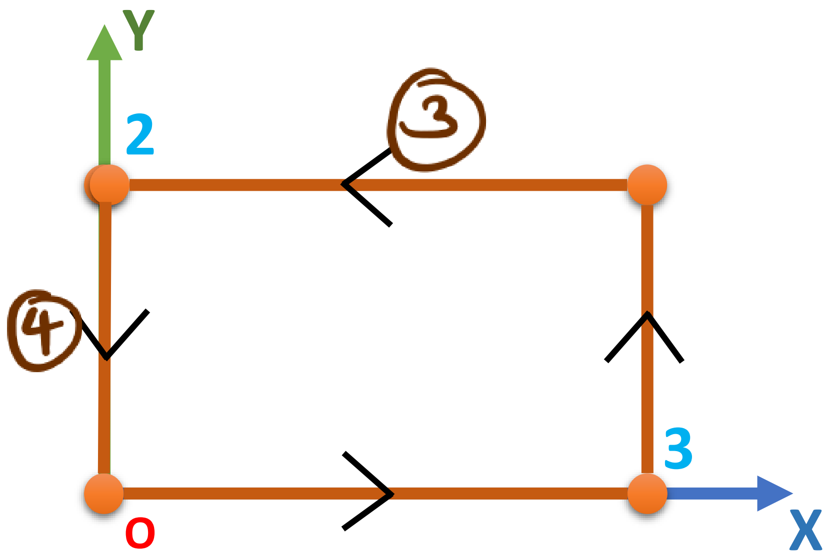
$= \left(3(y) + y^3/3 \right)_0^2$

$= 6 + 8/3$

$= 26/3$

Engineering Funda

$$\vec{F} = x^2y\,\hat{i} + (x+y^2)\,\hat{j} + xy\,\hat{k}$$



~ $d\vec{l} = dx\,\hat{i}$, $x \in (3,0)$, $y = 2$

~ $\int_{3} \vec{F}\,d\vec{l} = \int_{3}^{0} x^2 y\,dx$

$= 2\left(\frac{x^3}{3}\right)_3^0$

$= -18$

~ $d\vec{l} = dy\,\hat{j}$, $y \in (2,0)$, $x = 0$

~ $\int_{4} \vec{F}\,d\vec{l} = \int_{2}^{0} (x+y^2)\,dy$

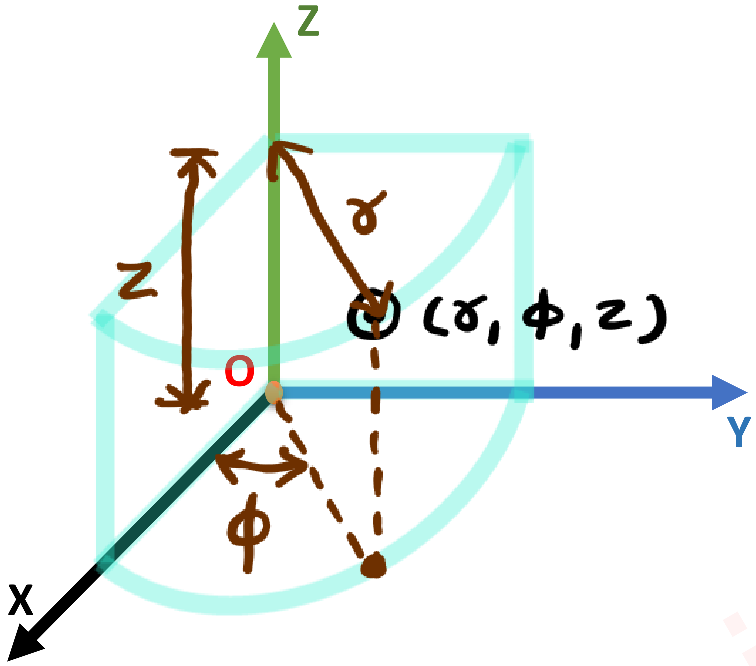
$= \left(\frac{y^3}{3}\right)_2^0$

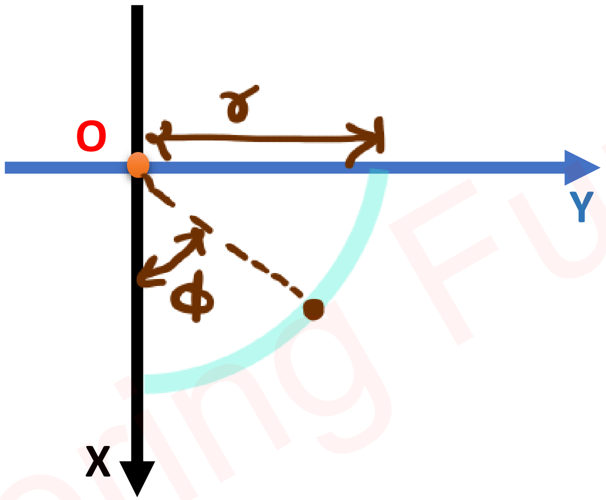
$= -8/3$

~ $\int \vec{F}\,d\vec{l} = -8/3 - 18 + 26/3 + 0 = \underline{\underline{-12}}$

Engineering Funda

Cylindrical Coordinate System



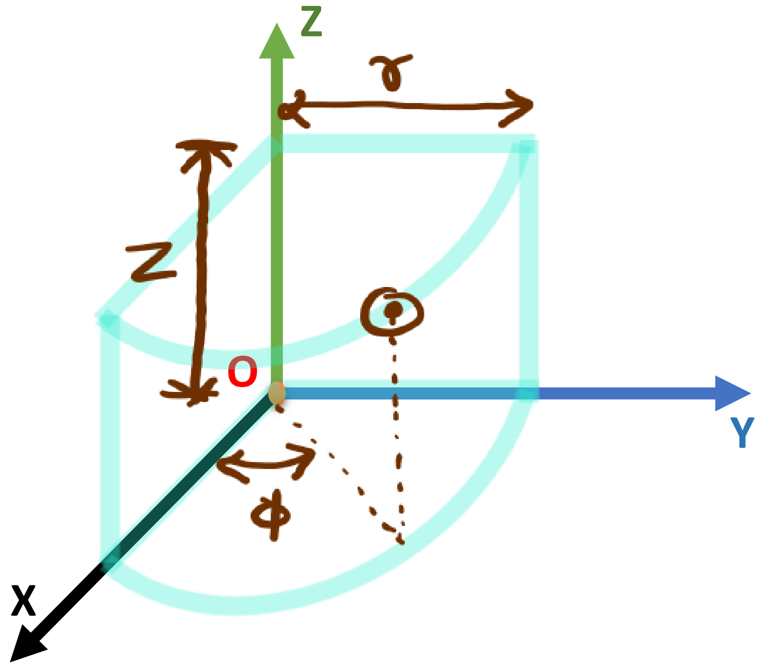


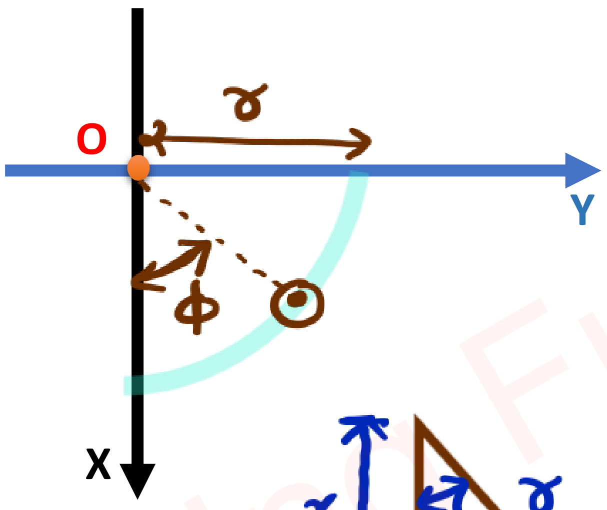
→ (r, ϕ, z)

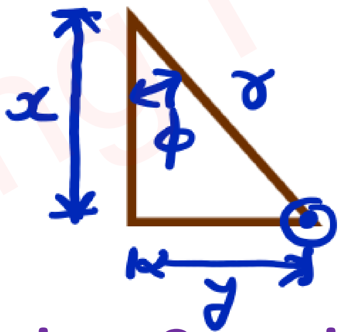
- z → Length of Cylinder
- ϕ → Angle on xy plane w.r.t x Axis.
- r → Radius of Cylinder.

Engineering Funda

Conversion of Cylindrical Coordinate into Cartesian Coordinate







$\rightarrow (r, \phi, z) \rightarrow (x, y, z)$

$\rightsquigarrow x = r\cos\phi$

$y = r\sin\phi$

$z = z$

❑ Convert Cylindrical Coordinate $(5, 60^0, 2)$ unit into Cartesian Coordinate (x, y, z).

$\rightsquigarrow \quad x = r\cos\phi = 5\cos 60^\circ = 5/2$

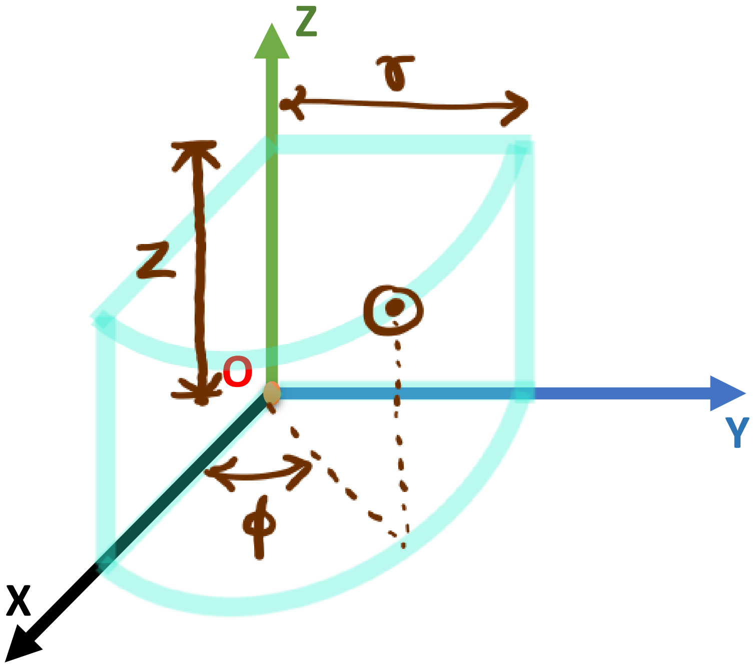
$y = r\sin\phi = 5\sin 60^\circ = 5\sqrt{3}/2$

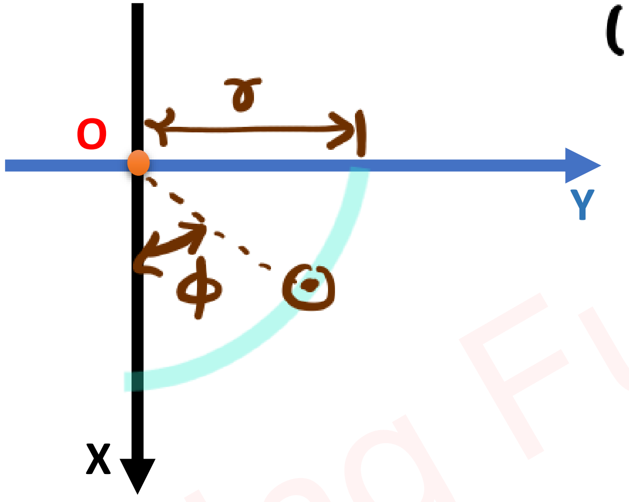
$z = 2$

$= \left(5/2, 5\sqrt{3}/2, 2\right)$

Engineering Funda

Conversion of Cartesian Coordinate into Cylindrical Coordinate





$$(x, y, z) \longrightarrow (r, \phi, z)$$

$$\Rightarrow x^2 + y^2 = r^2\cos^2\phi + r^2\sin^2\phi$$

$$\Rightarrow x^2 + y^2 = r^2$$

$$\Rightarrow r = \sqrt{x^2 + y^2}$$

$$\Rightarrow \frac{y}{x} = \frac{r\sin\phi}{r\cos\phi} = \tan\phi$$

$$\Rightarrow \phi = \tan^{-1}(y/x)$$

❑ Convert Cartesian Coordinate $(1, 1, 2)$ unit into Cylindrical Coordinate (r, ϕ, z).

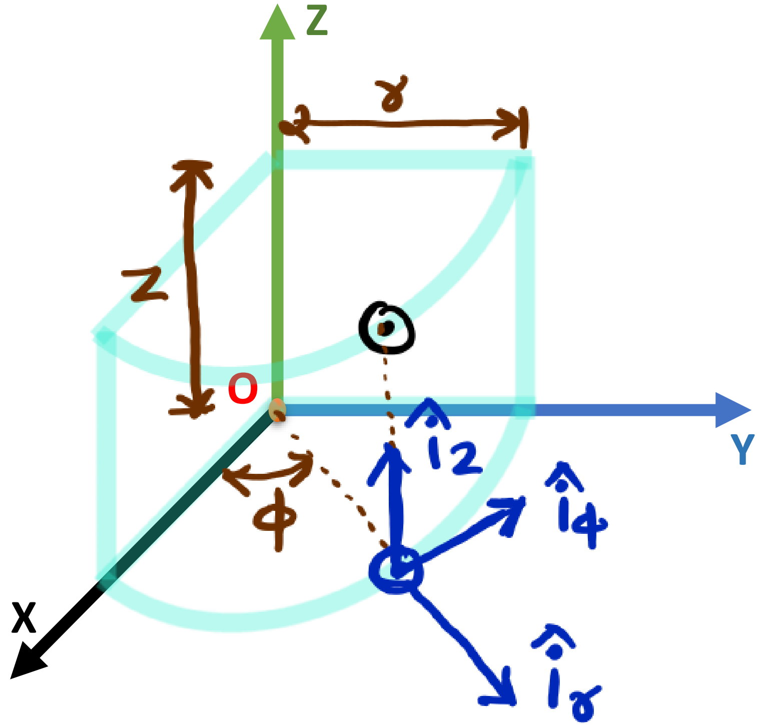
$$\Rightarrow r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

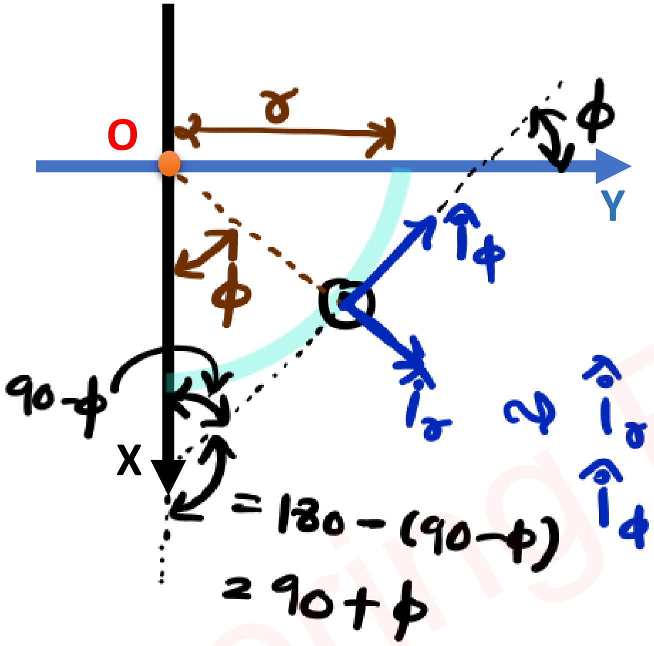
$$\phi = \tan^{-1}(y/x) = 45^\circ$$

$$z = 2$$

Engineering Funda

Conversion of Cartesian Vector into Cylindrical Vector (r, ϕ, z)





→ $\vec{P} = a_x \hat{i}_x + a_y \hat{i}_y + a_z \hat{i}_z$

$$\Downarrow$$

$\vec{P} = a_r \hat{i}_r + a_\phi \hat{i}_\phi + a_z \hat{i}_z$

→ $\hat{i}_r$ - It is in radial direction

$\hat{i}_\phi$ - It is tangent to curve from x to y direction.

→ $a_r = \vec{P} \cdot \hat{i}_r = (a_x \hat{i}_x + a_y \hat{i}_y + a_z \hat{i}_z) \cdot \hat{i}_r$

$= a_x (\underbrace{\hat{i}_x \cdot \hat{i}_r}_{\phi}) + a_y (\underbrace{\hat{i}_y \cdot \hat{i}_r}_{90-\phi}) + a_z (\underbrace{\hat{i}_z \cdot \hat{i}_r}_{90})$

$= a_x \cos\phi + a_y \sin\phi$

→ $a_\phi = \vec{P} \cdot \hat{i}_\phi = (a_x \hat{i}_x + a_y \hat{i}_y + a_z \hat{i}_z) \cdot \hat{i}_\phi$

$= a_x (\underbrace{\hat{i}_x \cdot \hat{i}_\phi}_{90+\phi}) + a_y (\underbrace{\hat{i}_y \cdot \hat{i}_\phi}_{\phi}) + a_z (\underbrace{\hat{i}_z \cdot \hat{i}_\phi}_{90})$

$= -a_x \sin\phi + a_y \cos\phi$

❑ Convert Cartesian Vector $\left(\vec{F} = 2\sqrt{3}\,\hat{\imath} + 2\hat{\jmath} + 4\hat{k}\right)$ into Cylindrical Vector.

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{2}{2\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$$

$$a_r = a_x \cos\phi + a_y \sin\phi$$
$$= 2\sqrt{3}\cos 30 + 2 \sin 30$$
$$= 2\sqrt{3}\left(\frac{\sqrt{3}}{2}\right) + 2\left(\frac{1}{2}\right)$$
$$= 4$$

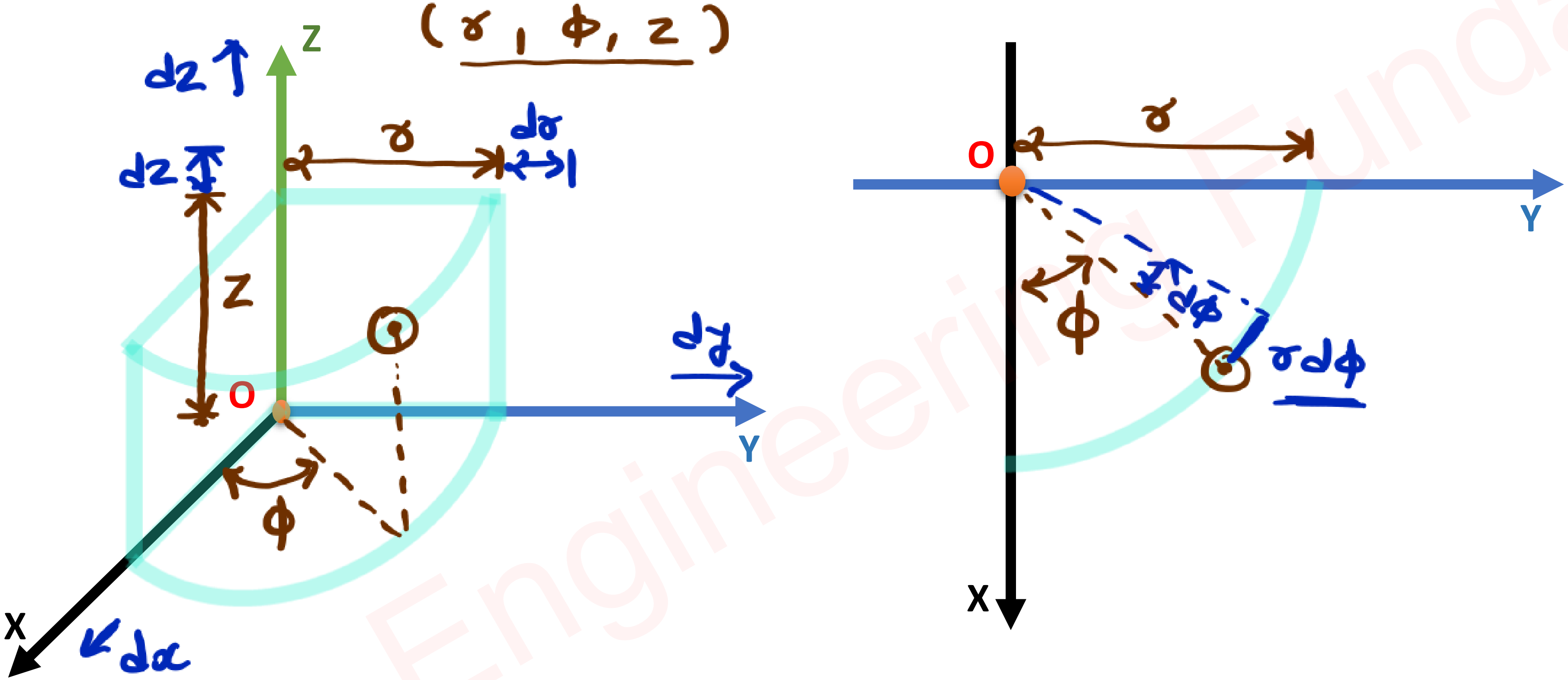
$$a_\phi = -a_x \sin\phi + a_y \cos\phi$$
$$= -2\sqrt{3}\sin 30 + 2\cos 30$$
$$= -\cancel{2}\sqrt{3}\left(\frac{1}{\cancel{2}}\right) + \cancel{2}\left(\frac{\sqrt{3}}{\cancel{2}}\right)$$
$$= 0$$

$$a_z = 4$$

$$\vec{F} = a_r \hat{i}_r + a_\phi \hat{i}_\phi + a_z \hat{i}_z$$
$$= 4\hat{i}_r + 4\hat{i}_z$$

Engineering Funda

Parameters of Cylindrical Coordinate System



Cartesian Coordinate System

Change in Coordinates

$\rightsquigarrow dx,\ dy,\ dz$

Line Integration

$\rightsquigarrow d\vec{l} = dx\,\hat{i}_x + dy\,\hat{i}_y + dz\,\hat{i}_z$

Surface Integration

$\rightsquigarrow d\vec{s} = dx\,dy\,\hat{i}_z$

$= dx\,dz\,\hat{i}_y$

$= dy\,dz\,\hat{i}_x$

Volume Integration

$\rightsquigarrow dV = dx\,dy\,dz$

Del Operator {Gradient}

$\rightsquigarrow \vec{\nabla} F = \left(\frac{\partial}{\partial x}\hat{i}_x + \frac{\partial}{\partial y}\hat{i}_y + \frac{\partial}{\partial z}\hat{i}_z\right) F$

Cylindrical Coordinate System

Change in Coordinates:

$\rightsquigarrow dr,\ r\,d\phi,\ dz$

Line Integration:

$\rightsquigarrow d\vec{l} = dr\,\hat{i}_r + r\,d\phi\,\hat{i}_\phi + dz\,\hat{i}_z$

Surface Integration:

$\rightsquigarrow d\vec{s} = r\,dr\,d\phi\,\hat{i}_z$

$= dr\,dz\,\hat{i}_\phi$

$= r\,d\phi\,dz\,\hat{i}_r$

Volume Integration:

$\rightsquigarrow dV = r\,dr\,d\phi\,dz$

Del Operator {Gradient}:

$\rightsquigarrow \vec{\nabla} F = \left(\frac{\partial}{\partial r}\hat{i}_r + \frac{\partial}{r\partial \phi}\hat{i}_\phi + \frac{\partial}{\partial z}\hat{i}_z\right) F$

Engineering Funda

Cartesian Coordinate System

Curl of Function

$$\vec{F} = F_x\widehat{i_x} + F_y\widehat{i_y} + F_z\widehat{i_z}$$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i}_x & \hat{i}_y & \hat{i}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

Divergence of Function

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Cylindrical Coordinate System

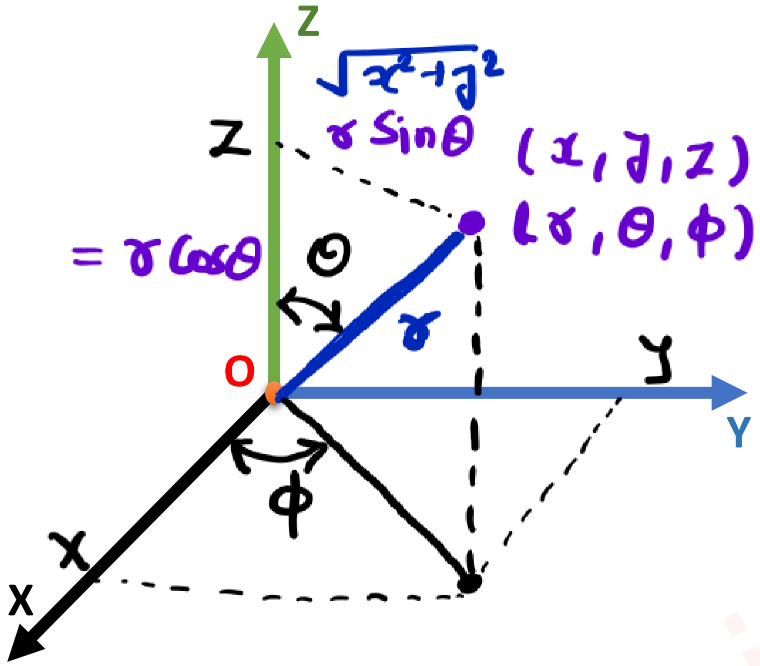
$$\vec{F} = F_r\widehat{i_r} + F_\phi\widehat{i_\phi} + F_z\widehat{i_z}$$

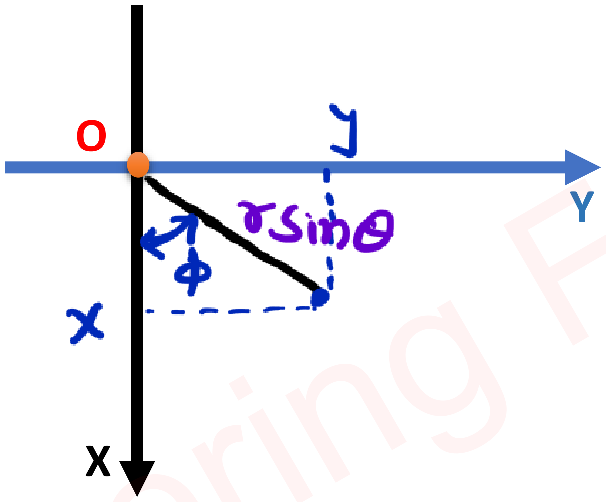
$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \left(\frac{1}{r}\right)\hat{i}_r & \hat{i}_\phi & \left(\frac{1}{r}\right)\hat{i}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_r & rF_\phi & F_z \end{vmatrix}$$

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r}\frac{\partial (rF_r)}{\partial r} + \frac{1}{r}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

Engineering Funda

Spherical Coordinate System





$[r, \theta, \phi]$

- r → Radius
- θ → Angle w.r.t to z Axis
- ϕ → Angle in xy plane w.r.t to x Axis

$z = r\cos\theta$

$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$r = \sqrt{x^2+y^2+z^2}$

$\phi = \tan^{-1}(y/x)$

$\theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)$

Engineering Funda

Conversion of Spherical Coordinate into Cartesian Coordinate

❑ Convert Spherical Coordinate $(4, 30^0, 60^0)$ unit into Cartesian Coordinate (x, y, z).

$\rightsquigarrow$ $x = r \sin\theta \cos\phi = 4 \sin 30^\circ \cos 60^\circ = 1$

$y = r \sin\theta \sin\phi = 4 \sin 30 \sin 60 = \sqrt{3}$

$z = r \cos\theta = 4 \cos 30 = 2\sqrt{3}$

Conversion of Cartesian Coordinate into Spherical Coordinate

❑ Convert Cartesian Coordinate $(1, 1, 2)$ unit into Spherical Coordinate (r, θ, ϕ).

$\rightsquigarrow$ $r = \sqrt{1^2 + 1^2 + 2^2} = \sqrt{6}$

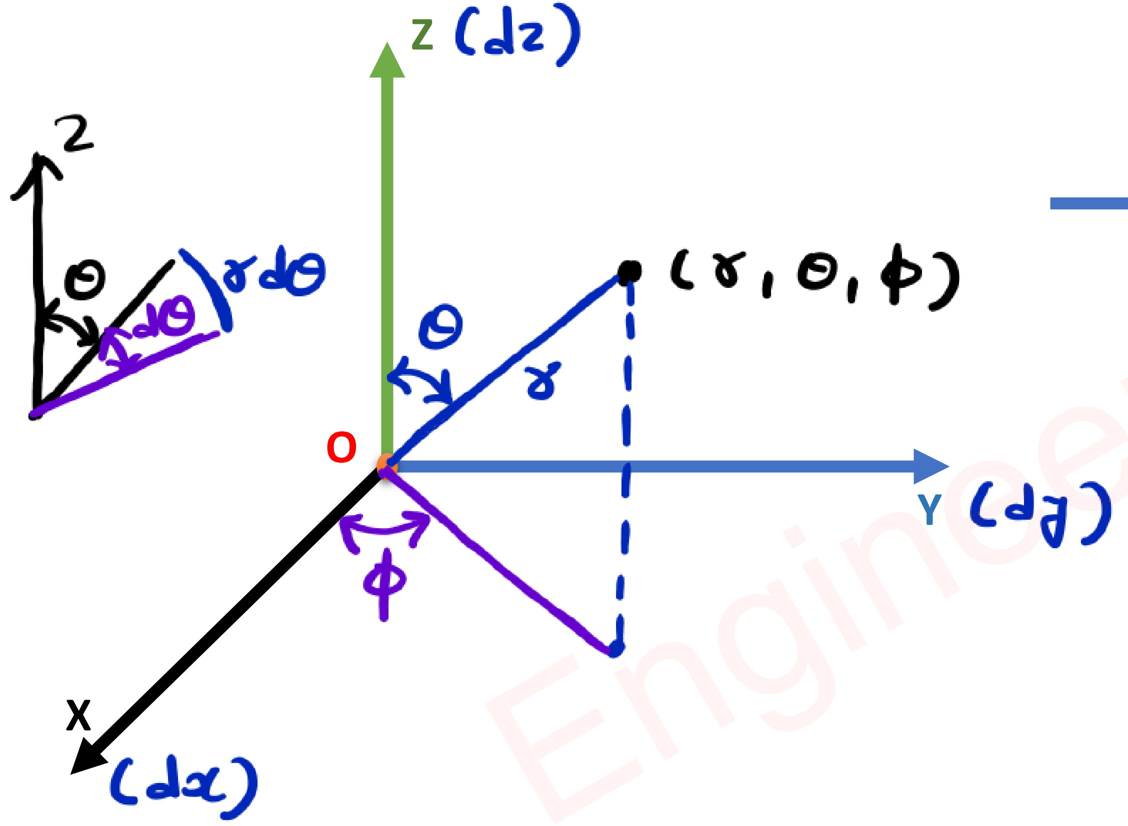
$\phi = \tan^{-1}\left(y/x\right) = \tan^{-1}\left(1/1\right) = 45^\circ$

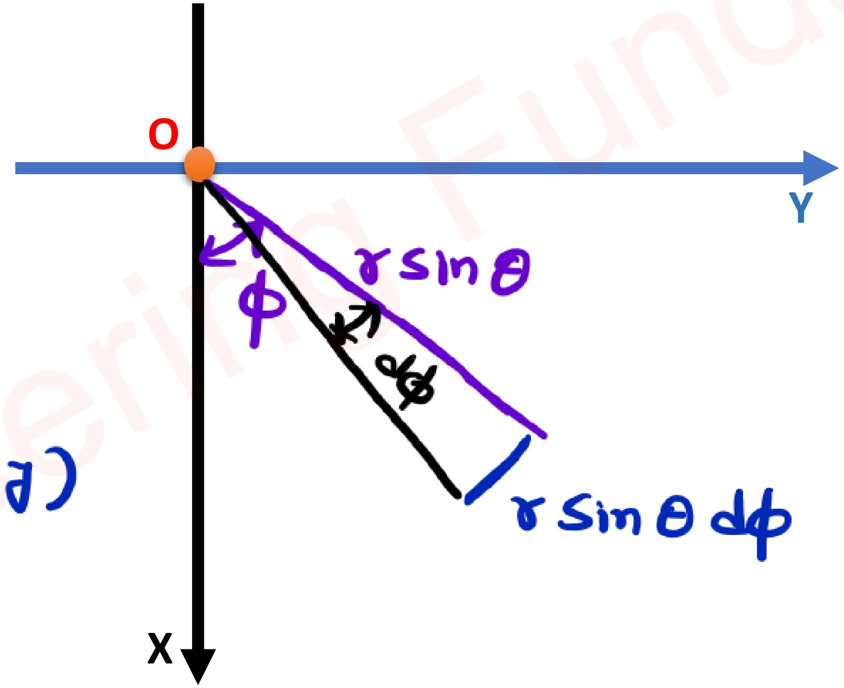
$\theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right) = \tan^{-1}\left(\frac{\sqrt{2}}{2}\right) = 35^\circ$

$= (\sqrt{6}, 35^\circ, 45^\circ)$

Engineering Funda

Parameters of Spherical Coordinate System





$\rightsquigarrow\ dr,\ r\,d\theta\ ,\ r\sin\theta\ d\phi$

Engineering Funda

Cartesian Coordinate System	Spherical Coordinate System
Change in Coordinates	
$\rightsquigarrow dx,\ dy,\ dz$	$\rightsquigarrow dr,\ r\,d\theta,\ r\sin\theta\,d\phi$
Line Integration	
$\rightsquigarrow d\vec{l} = dx\,\hat{i}_x + dy\,\hat{i}_y + dz\,\hat{i}_z$	$\rightsquigarrow d\vec{l} = dr\,\hat{i}_r + r\,d\theta\,\hat{i}_\theta + r\sin\theta\,d\phi\,\hat{i}_\phi$
Surface Integration	
$\rightsquigarrow d\vec{S} = dx\,dy\,\hat{i}_z$ $= dx\,dz\,\hat{i}_y$ $= dy\,dz\,\hat{i}_x$	$\rightsquigarrow d\vec{S} = r\,dr\,d\theta\,\hat{i}_\phi$ $= r\sin\theta\,dr\,d\phi\,\hat{i}_\theta$ $= r^2\sin\theta\,d\theta\,d\phi\,\hat{i}_r$
Volume Integration	
$\rightsquigarrow dv = dx\,dy\,dz$	$\rightsquigarrow dv = r^2\sin\theta\,dr\,d\theta\,d\phi$
Del Operator {Gradient}	
$\rightsquigarrow \vec{\nabla} = \frac{\partial}{\partial x}\hat{i}_x + \frac{\partial}{\partial y}\hat{i}_y + \frac{\partial}{\partial z}\hat{i}_z$	$\rightsquigarrow \vec{\nabla} = \frac{\partial}{\partial r}\hat{i}_r + \frac{\partial}{r\,\partial\theta}\hat{i}_\theta + \frac{\partial}{r\sin\theta\,\partial\phi}\hat{i}_\phi$

Engineering Funda

Cartesian Coordinate System

❑ Curl of Function

$$\vec{F} = F_x\hat{i_x} + F_y\hat{i_y} + F_z\hat{i_z}$$

$$\rightsquigarrow \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i}_x & \hat{i}_y & \hat{i}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

❑ Divergence of Function

$$\rightsquigarrow \vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Spherical Coordinate System

$$\vec{F} = F_r\hat{i_r} + F_\theta\hat{i_\theta} + F_\phi\hat{i_\phi}$$

$$\rightsquigarrow \vec{\nabla} \times \vec{F} = \begin{vmatrix} \left(\frac{1}{r^2 \sin\theta}\right)\hat{i}_r & \left(\frac{1}{r \sin\theta}\right)\hat{i}_\theta & \left(\frac{1}{r}\right)\hat{i}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & r F_\theta & r\sin\theta\, F_\phi \end{vmatrix}$$

$$\rightsquigarrow \vec{\nabla} \cdot \vec{F} = \frac{1}{r^2}\frac{\partial (r^2 F_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial (\sin\theta F_\theta)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial F_\phi}{\partial \phi}$$

Engineering Funda

Vector Conversion: Spherical to Cartesian & Visa versa

	$\hat{i}_r$	$\hat{i}_\theta$	$\hat{i}_\phi$
$\hat{i}_x$	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
$\hat{i}_y$	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
$\hat{i}_z$	$\cos\theta$	$-\sin\theta$	0

$$\hat{i}_r = \sin\theta\cos\phi\,\hat{i}_x + \sin\theta\sin\phi\,\hat{i}_y + \cos\theta\,\hat{i}_z$$

$$\hat{i}_\theta = \cos\theta\cos\phi\,\hat{i}_x + \cos\theta\sin\phi\,\hat{i}_y - \sin\theta\,\hat{i}_z$$

$$\hat{i}_\phi = -\sin\phi\,\hat{i}_x + \cos\phi\,\hat{i}_y$$

Engineering Funda

❑ Convert the given vector $\vec{A} = 10\hat{i_z}$ into Spherical vector at a Coordinate $(4, 30^0, 60^0)$.

(r, θ, ϕ)

$\rightsquigarrow \vec{A} = A_r \hat{i}_r + A_\theta \hat{i}_\theta + A_\phi \hat{i}_\phi$

$\rightsquigarrow A_r = \vec{A} \cdot \hat{i}_r = 10 \cos\theta = 5\sqrt{3}$

$\rightsquigarrow A_\theta = \vec{A} \cdot \hat{i}_\theta = -10 \sin\theta = -5$

$\rightsquigarrow A_\phi = \vec{A} \cdot \hat{i}_\phi = 0$

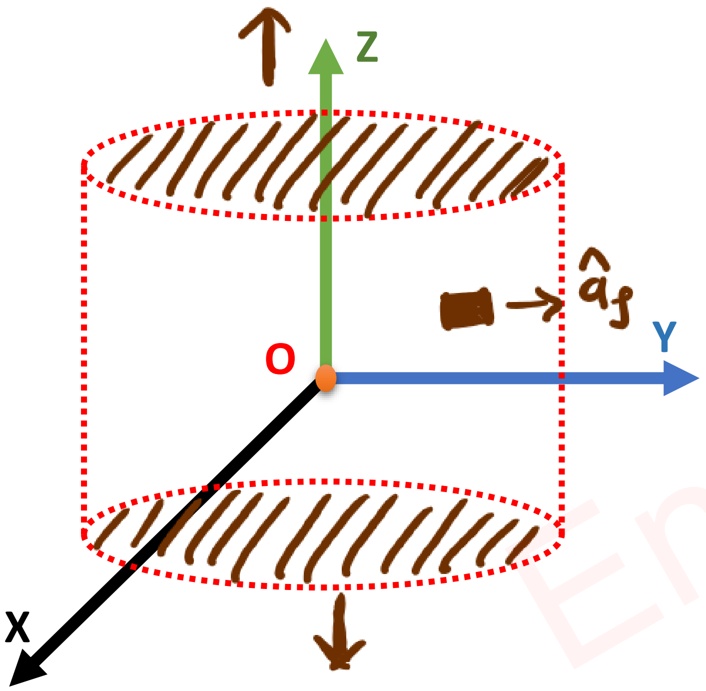
$\boxed{\rightsquigarrow \vec{A} = 5\sqrt{3}\, \hat{i}_r - 5\, \hat{i}_\theta}$

Engineering Funda

Examples of Coordinate System

Question 1 – If $\vec{D} = 2\rho z^2\ \widehat{a_\rho} + \rho\cos^2\Phi\ \widehat{a_z}$, the region defined by $0 \le \rho \le 5$, $-1 \le z \le 1$, and $0 \le \Phi \le 2\pi$, as shown in figure.

The value of $\oint \vec{D}\,.\,\overrightarrow{dS}$ is …$200\pi/3$…



$$\oint \vec{D}\cdot d\vec{S} = \underbrace{\int \vec{D}\cdot d\vec{S}}_{\text{Top}} + \underbrace{\int \vec{D}\cdot d\vec{S}}_{\text{Bottom}} + \underbrace{\int \vec{D}\cdot d\vec{S}}_{\text{Curved}}$$

$$= \int \rho\cos^2\phi\,(\rho\,d\rho\,d\phi) + \int -\rho\cos^2\phi\,(\rho\,d\rho\,d\phi) + \int_{-1}^{1}\int_{0}^{2\pi} 2\rho z^2\,(\rho\,d\phi\,dz)$$

$$= 2\rho^2\,[\phi]_0^{2\pi}\left[\frac{z^3}{3}\right]_{-1}^{1}$$

$$= 25 \times 2 \times 2\pi \times \left(\frac{2}{3}\right)$$

$$= 200\pi/3$$

Engineering Funda

Question 2 – Given a conservative field, If $\overrightarrow{M} = [z\cos(xz) + y]\widehat{a_x} + 2Kx\widehat{a_y} + [x\cos(xz)]\widehat{a_z}$, The value of K is ....$\underline{\underline{1/2}}$..................

$$\rightsquigarrow \vec{\nabla} \times \vec{m} = 0 = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z\cos(xz)+y & 2Kx & x\cos(xz) \end{vmatrix}$$

$$\Rightarrow 0 = \hat{a}_x(0-0) - \hat{a}_y\left(\frac{\partial(x\cos(xz))}{\partial x} - \frac{\partial(z\cos(xz)+y)}{\partial z}\right) +$$

$$\hat{a}_z\left(\frac{\partial(2Kx)}{\partial x} - \frac{\partial(z\cos(xz)+y)}{\partial y}\right)$$

$$\Rightarrow 0 = \hat{a}_y(\cancel{\cos(xz)} + xz(\cancel{-\sin(xz)}) - \cancel{\cos(xz)} + zx\,\cancel{\sin(xz)}) +$$

$$\hat{a}_z(2K-1)$$

$$\Rightarrow 2K-1 = 0$$

$$\Rightarrow K = 1/2$$

Engineering Funda

Question 3 – **A scalar field g will be harmonic at all the points for the value of K ……$-\frac{1}{2}$……**
Where, $g = (1 + 2K)x^2y + xyz$

$\Rightarrow \nabla^2 g = 0 = \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2}$

$\Rightarrow (1+2K)\,2y = 0$

$\Rightarrow K = -\frac{1}{2}$

Engineering Funda

Examples of Coordinate System

Question 4 – Given $\vec{A} = (x + 4z)\widehat{a_x} + (2x - 3y)\widehat{a_y} + (4x + 3y - Cz)\widehat{a_z}$,

The value of C for which $\vec{A}$ is solenoid will be ....$\underline{\underline{-2}}$..................

$\Rightarrow$ For Solenoid, $\nabla \cdot \vec{A} = 0$

$$\Rightarrow \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0$$

$$\Rightarrow 1 - 3 - c = 0$$

$$\Rightarrow \boxed{c = -2}$$

Engineering Funda

Question 5 – The direction of $\vec{A}$ is radially outward from the origin with $|\vec{A}| = Kr^n$, $r^2 = x^2 + y^2 + z^2$ and K is constant. The value of n for which $\nabla.\vec{A} = 0$ will be ......-2..........

$\rightsquigarrow \vec{A} = |\vec{A}|\ \hat{a}_r$ ← Spherical Coordinates.

$= Kr^n\ \hat{a}_r$

$\rightsquigarrow \nabla.\vec{A} = 0$

$\Rightarrow \frac{1}{r^2}\frac{\partial(r^2 A_r)}{\partial r} = 0$

$\Rightarrow \frac{1}{r^2}\frac{\partial(Kr^{n+2})}{\partial r} = 0$

$\Rightarrow \frac{1}{r^2}(K(n+2)r^{n+1}) = 0$

$\Rightarrow n = -2$

Engineering Funda

Question 6 – For a vector $\vec{A} = (4x + K_1 z)\widehat{a_x} + (K_2 x - 5z)\widehat{a_y} + (4x - K_3 y + 2z)\widehat{a_z}$ to be irrotational, the value of K_1, K_2 and K_3 are ....

$\Rightarrow \nabla \times \vec{A} = 0$ ← For irrotational vector.

$$\Rightarrow \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (4x + K_1 z) & (K_2 x - 5z) & (4x - K_3 y + 2z) \end{vmatrix} = 0$$

$$\Rightarrow \hat{a}_x(-K_3 + 5) - \hat{a}_y(4 - K_1) + \hat{a}_z(K_2 - 0) = 0$$

$\Rightarrow K_3 = 5$, $K_1 = 4$, $K_2 = 0$

Engineering Funda

Examples of Coordinate System

Question 7 – If the vectors $\vec{A}$ and $\vec{B}$ are conservative, then

a. $\vec{A} \times \vec{B}$ is solenoidal ✓
b. $\vec{A} \times \vec{B}$ is conservative
c. $\vec{A} + \vec{B}$ is solenoidal
d. $\vec{A} - \vec{B}$ is solenoidal

$$\left.\begin{aligned} \nabla \times \vec{A} &= 0 \\ \nabla \times \vec{B} &= 0 \end{aligned}\right\} \text{Conservative Vectors}$$

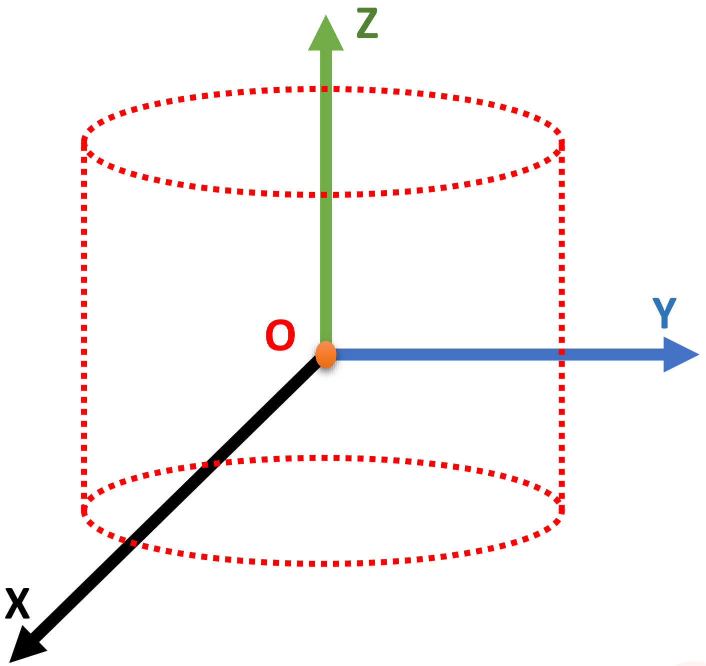
$$\begin{aligned} \nabla \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B}) \\ &= 0 - 0 \\ &= 0 \end{aligned}$$

Hence, $\vec{A} \times \vec{B}$ is Solenoidal.

Engineering Funda

Question 8 – If $\vec{D} = 2\rho z^2\ \widehat{a_\rho} + \rho\cos^2\Phi\ \widehat{a_z}$, the region defined by $0 \le \rho \le 5$, $-1 \le z \le 1$, and $0 \le \Phi \le 2\pi$, as shown in figure.

The value of $\oint \vec{D}\,.\,\vec{dS}$ is ……$\underline{\underline{200\pi/3}}$……



$$\int \vec{D}.\vec{ds} = \int (\nabla.\vec{D})\, dv$$

$$\nabla.\vec{D} = \frac{1}{\rho}\frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$= \frac{1}{\rho}\, 2z^2(2\rho) + \frac{1}{\rho}(0) + 0$$

$$= 4z^2$$

$$\int (\nabla.\vec{D}).\, dv = \int_{-1}^{1}\int_{0}^{2\pi}\int_{0}^{5} 4z^2\, \rho\, d\rho\, d\phi\, dz$$

$$= 4\left(\frac{z^3}{3}\right)_{-1}^{1} \left(\frac{\rho^2}{2}\right)_{0}^{5} \left(\phi\right)_{0}^{2\pi}$$

$$= 4 \times \frac{2}{3} \times \frac{25}{2} \times 2\pi = \frac{200\pi}{3}$$

Question 9 – Find the distance between points P1 (1, 2, 3) and P2 (2, 30^0, 4).

$\rightsquigarrow P_1 = (1, 2, 3)$

$P_2 = (2, 30^\circ, 4) = (\sqrt{3}, 1, 4)$

$\downarrow r \quad \downarrow \phi \quad \downarrow z$

$\rightsquigarrow x = r \cos\phi = 2\cos 30^\circ = \sqrt{3}$

$y = r \sin\phi = 2 \sin 30^\circ = 1$

$z = 4$

$\rightsquigarrow d = \sqrt{(1-\sqrt{3})^2 + (2-1)^2 + (3-4)^2}$

$= 1.59$

Engineering Funda

Examples of Coordinate System

Question 10 – For a vector field $\vec{D} = (\rho \cos^2\phi)\widehat{a_\rho} + (z^2 \sin^2\phi)\widehat{a_\phi}$ in a cylindrical coordinate system (ρ, ϕ, z). The net flux (in coulomb) leaving the closed surface of the cylinder $(\rho = 3,\ 0 \le z \le 2)$ is ....

$$\oint \vec{D}\, d\vec{s} = \int (\nabla \cdot \vec{D})\, dv$$

$$\nabla \cdot \vec{D} = \frac{1}{\rho}\frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$= \frac{1}{\cancel{\rho}}((2\cancel{\rho})\cos^2\phi) + \frac{1}{\rho}(z^2(2\sin\phi\cos\phi)) + 0$$

$$= (1 + \cos 2\phi) + \frac{z^2}{\rho}\sin 2\phi$$

$$\int (\nabla \cdot \vec{D})\, dv = \int_0^2 \int_0^{2\pi} \int_0^3 \left((1 + \cos 2\phi) + \frac{z^2}{\rho}\sin 2\phi \right) \rho\, d\rho\, d\phi\, dz$$

Engineering Funda

$$= \int_0^2 \int_0^{2\pi} \int_0^3 \rho(1+\cos 2\phi) + z^2 \sin 2\phi \; d\rho \, d\phi \, dz$$

$$= \left[\frac{\rho^2}{2}\right]_0^3 \left[z\right]_0^2 \left[\phi + \frac{\sin 2\phi}{2}\right]_0^{2\pi} + \left[\rho\right]_0^3 \left[\frac{z^3}{3}\right]_0^2 \left[-\frac{\cos 2\phi}{2}\right]_0^{2\pi}$$

$$= \frac{9}{2} \times 2 \times 2\pi + (3)\left(\frac{8}{3}\right)(0)$$

$$\oint \vec{D} \cdot d\vec{s} = 18\pi \text{ Coloumb}$$

Engineering Funda

Question 11 – For a vector field A, which one of the following is false?

a. $\nabla \times \mathbf{A}$ is another vector field ✓

b. $\mathbf{A}$ is solenoidal, if $\nabla . \mathbf{A} = 0$ ✓

c. $\mathbf{A}$ is irrotational, if $\nabla^2 \mathbf{A} = 0$ ✗

d. $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla . \mathbf{A}) - \nabla^2 \mathbf{A}$ ✓

~ Cross product $\longrightarrow$ Vector

~ Irrotational $\longrightarrow$ $\nabla \times A = 0$

Engineering Funda

Question 12 – If vector $\vec{A}$ is orthogonal to $\vec{B}$ then $\vec{A} \times (\vec{B} \times \vec{C})$ is ……………………

→ If $\vec{A}$ & $\vec{B}$ are orthogonal ⇒ $\vec{A} \cdot \vec{B} = 0$

$$\Rightarrow \vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\underline{\vec{A} \cdot \vec{B}})$$

$$= \underline{\vec{B}}(\vec{A} \cdot \vec{C})$$

It is having direction of $\vec{B}$

Engineering Funda

Examples of Coordinate System

Question 13 – The surface $\rho = 3$, $\rho = 5$, $\phi = 100^0$, $\phi = 130^0$, $Z = 3$, and $Z = 4.5$ Define a closed volume. The length of the longest straight line that lies within volume is ....

$P_1(3, 100, 3) \longrightarrow (3\cos 100, 3\sin 100, 3) = (-0.52, 2.93, 3)$

$P_2(5, 130, 4.5) \longrightarrow (5\cos 130, 5\sin 130, 4.5) = (-3.21, 3.83, 4.5)$

$d = \sqrt{(-0.52+3.21)^2 + (2.93-3.83)^2 + (3-4.5)^2}$

$\boxed{d = 3.21}$

Engineering Funda

Question 14 – If $V = x^2y^2z^2$, The Laplacian of field V at point $(1, 45^0, 2)$ is ....

(r, ϕ, z)

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$= 2y^2z^2 + 2x^2z^2 + 2x^2y^2$$

$$= 2(x^2y^2 + x^2z^2 + y^2z^2)$$

$$P(1, 45^0, 2) \rightarrow (\cos 45, \sin 45, 2)$$

$$= (1/\sqrt{2}, 1/\sqrt{2}, 2)$$

$$\nabla^2 V \Big|_{(1/\sqrt{2}, 1/\sqrt{2}, 2)} = 2\left(\frac{1}{4} + 2 + 2\right)$$

$$= 8.5$$

$$= 17/2$$

Engineering Funda

Question 15 – If $V = xy - x^2y + y^2z^2$, The value of the Div Grad V at point $(1, 90^0, 1)$ is ..$\boxed{2}$..........

(r, ϕ, z)

$\rightsquigarrow$ Grad V $= \frac{\partial V}{\partial x}\hat{i}_x + \frac{\partial V}{\partial y}\hat{i}_y + \frac{\partial V}{\partial z}\hat{i}_z$

$= (y - 2xy)\hat{i}_x + (x - x^2 + 2yz^2)\hat{i}_y + (2zy^2)\hat{i}_z$

$\rightsquigarrow$ Div Grad V $= \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$

$= (-2y) + (2z^2) + 2y^2$

$\rightsquigarrow$ $P(1, 90^\circ, 1) \rightarrow (\cos 90, \sin 90, 1)$

$= (0, 1, 1)$

$\rightsquigarrow$ Div Grad V $\Big|_{(0,1,1)} = (-2) + 2 + 2$

$\boxed{= 2}$

Engineering Funda